

BFS Traversal

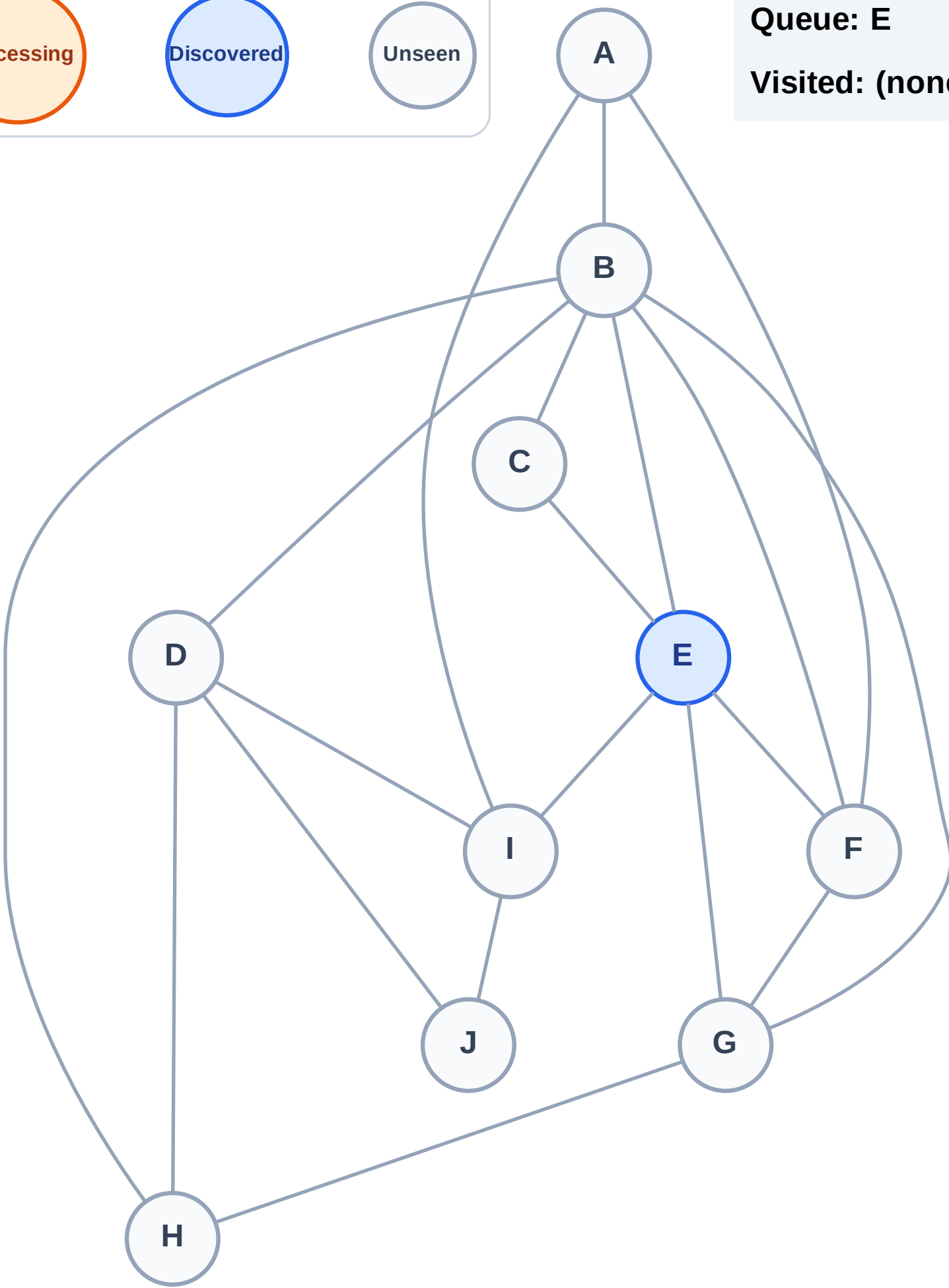
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

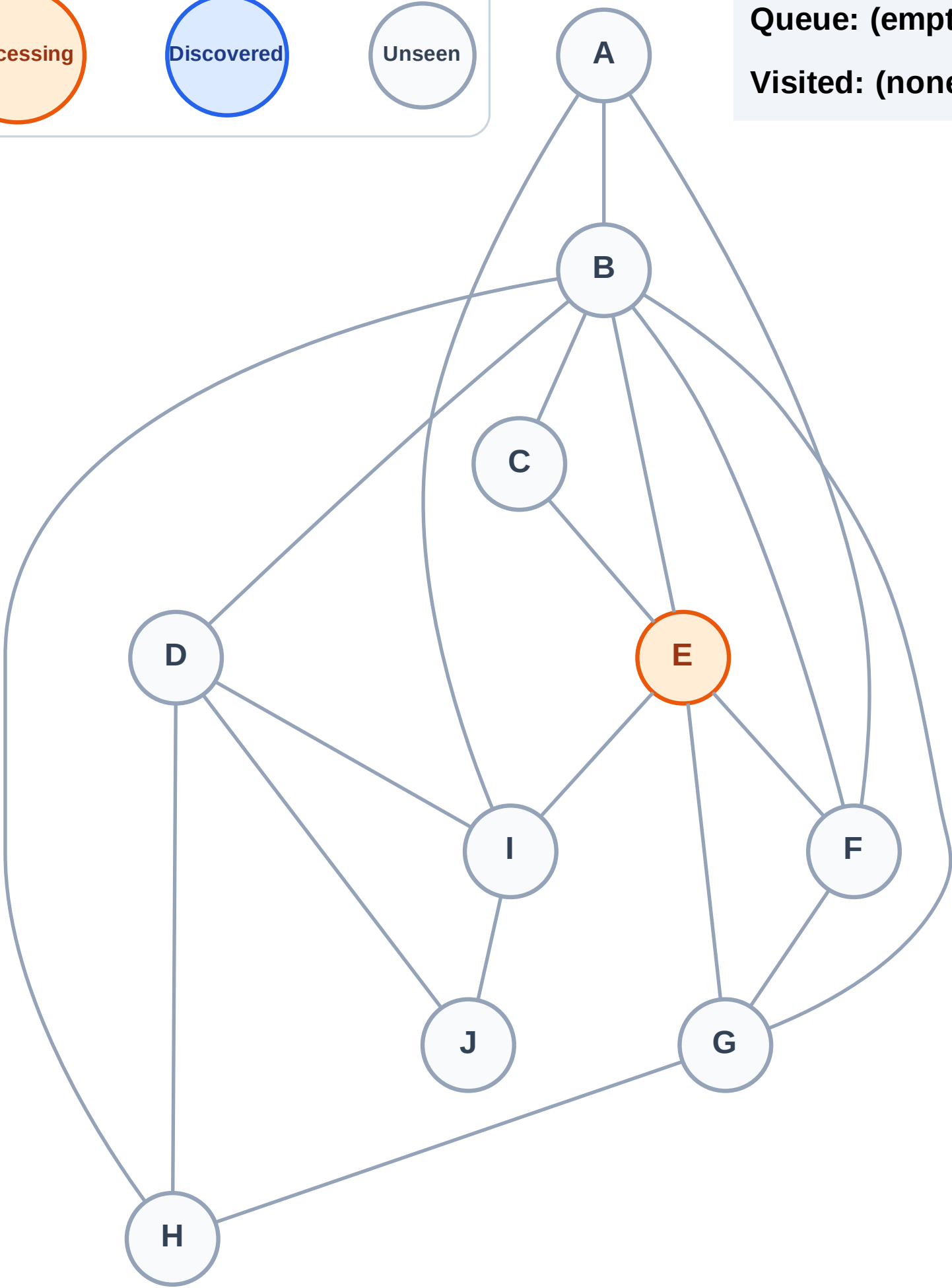
Step 2: Process node E

Legend



Queue: (empty)

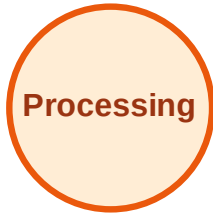
Visited: (none)



BFS Traversal

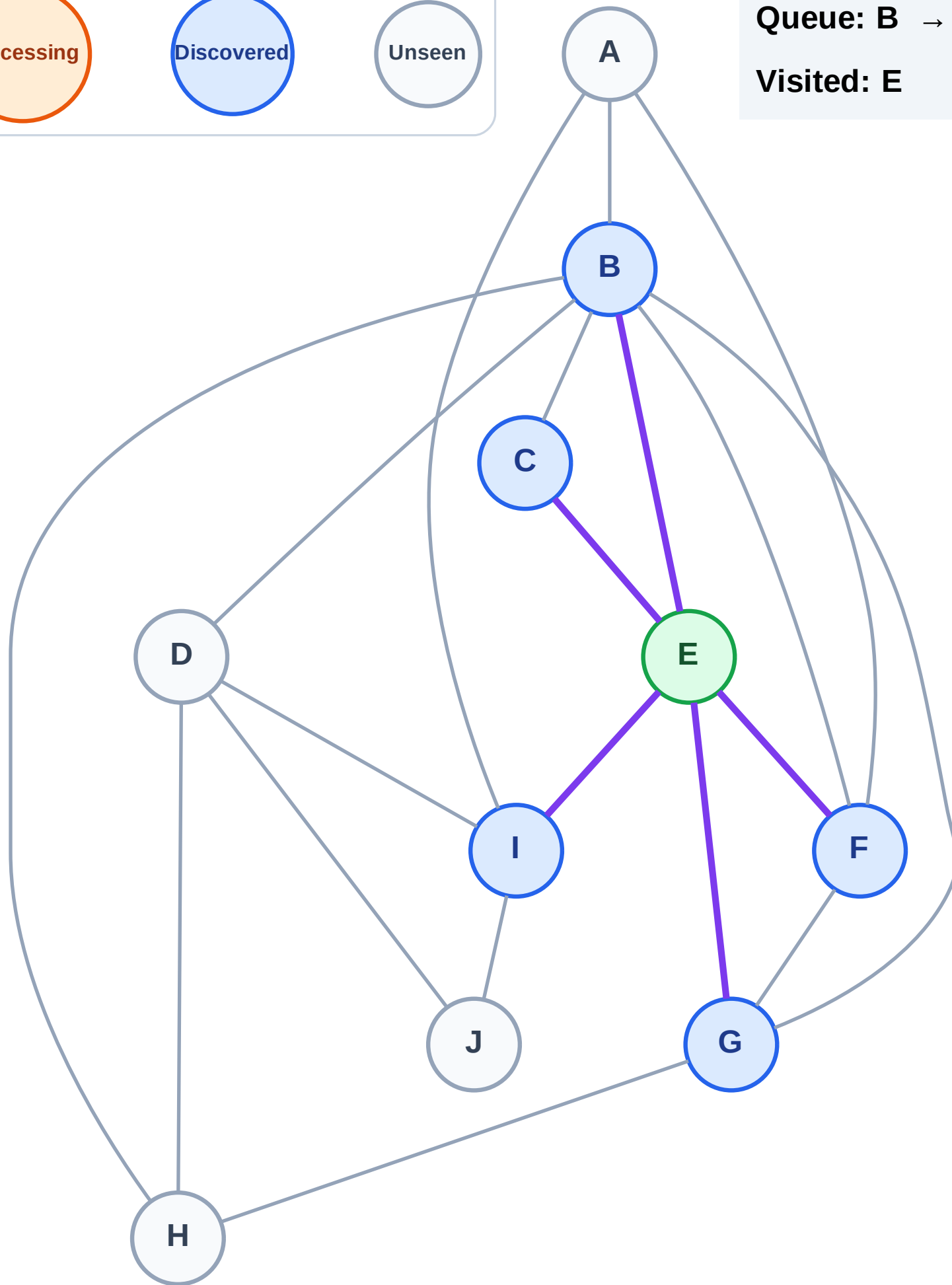
Step 3: Discover B, C, F, G, I from E

Legend



Queue: B → C → F → G → I

Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

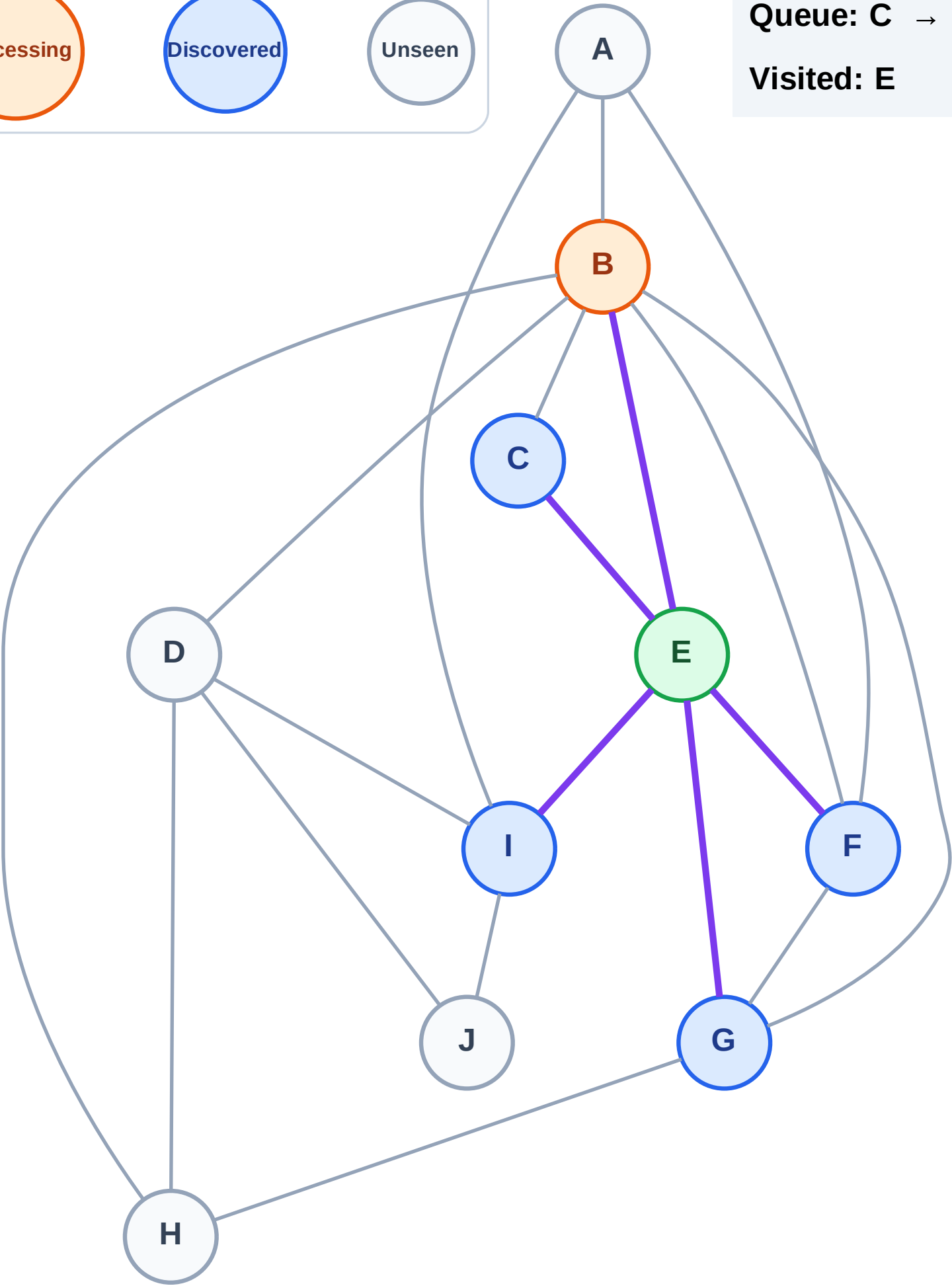
Processing

Discovered

Unseen

Queue: C → F → G → I

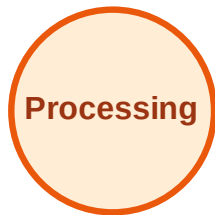
Visited: E



BFS Traversal

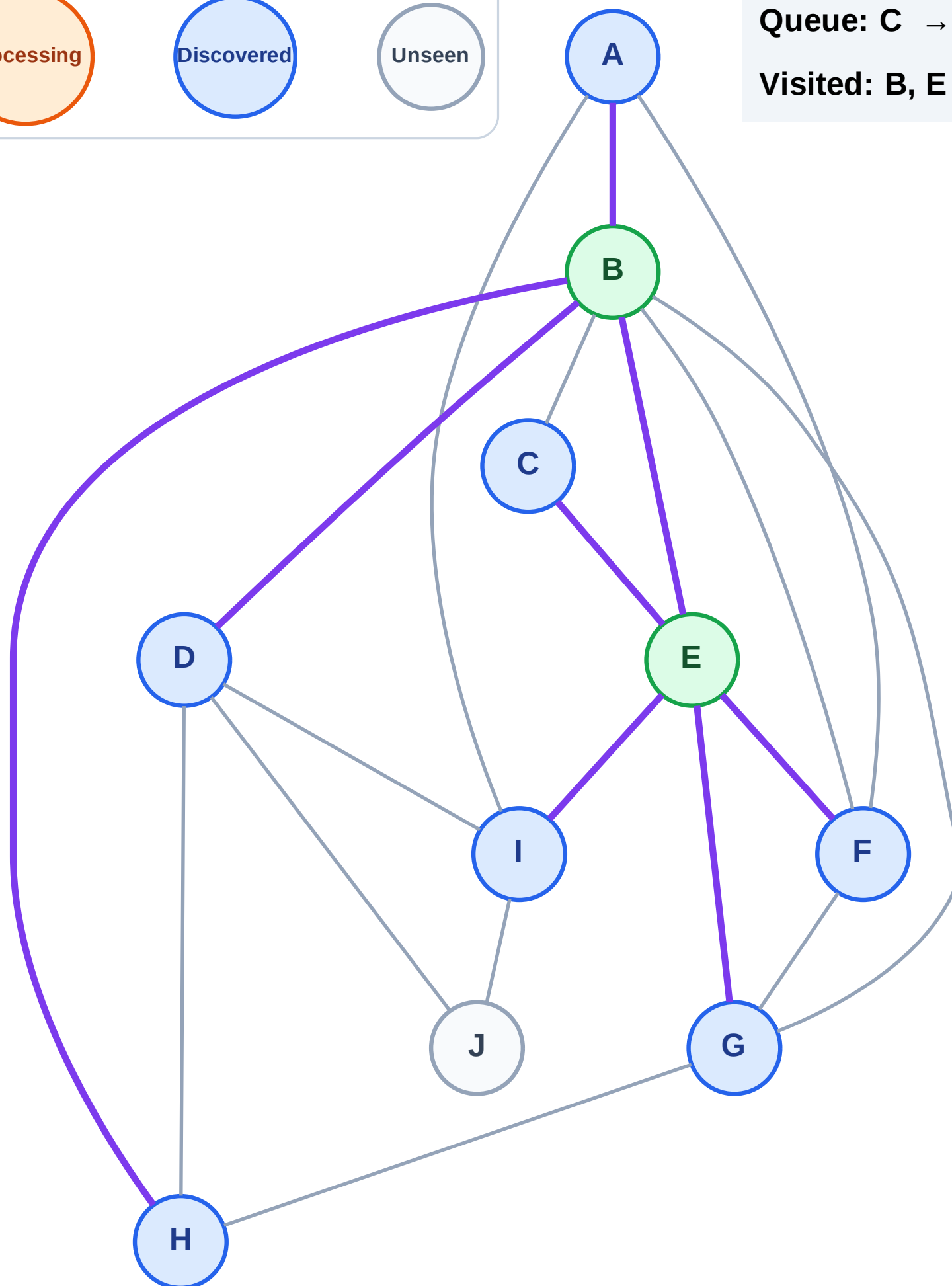
Step 5: Discover A, D, H from B

Legend



Queue: C → F → G → I → A → D → H

Visited: B, E



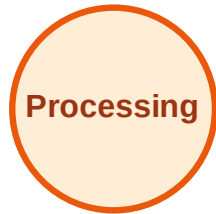
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



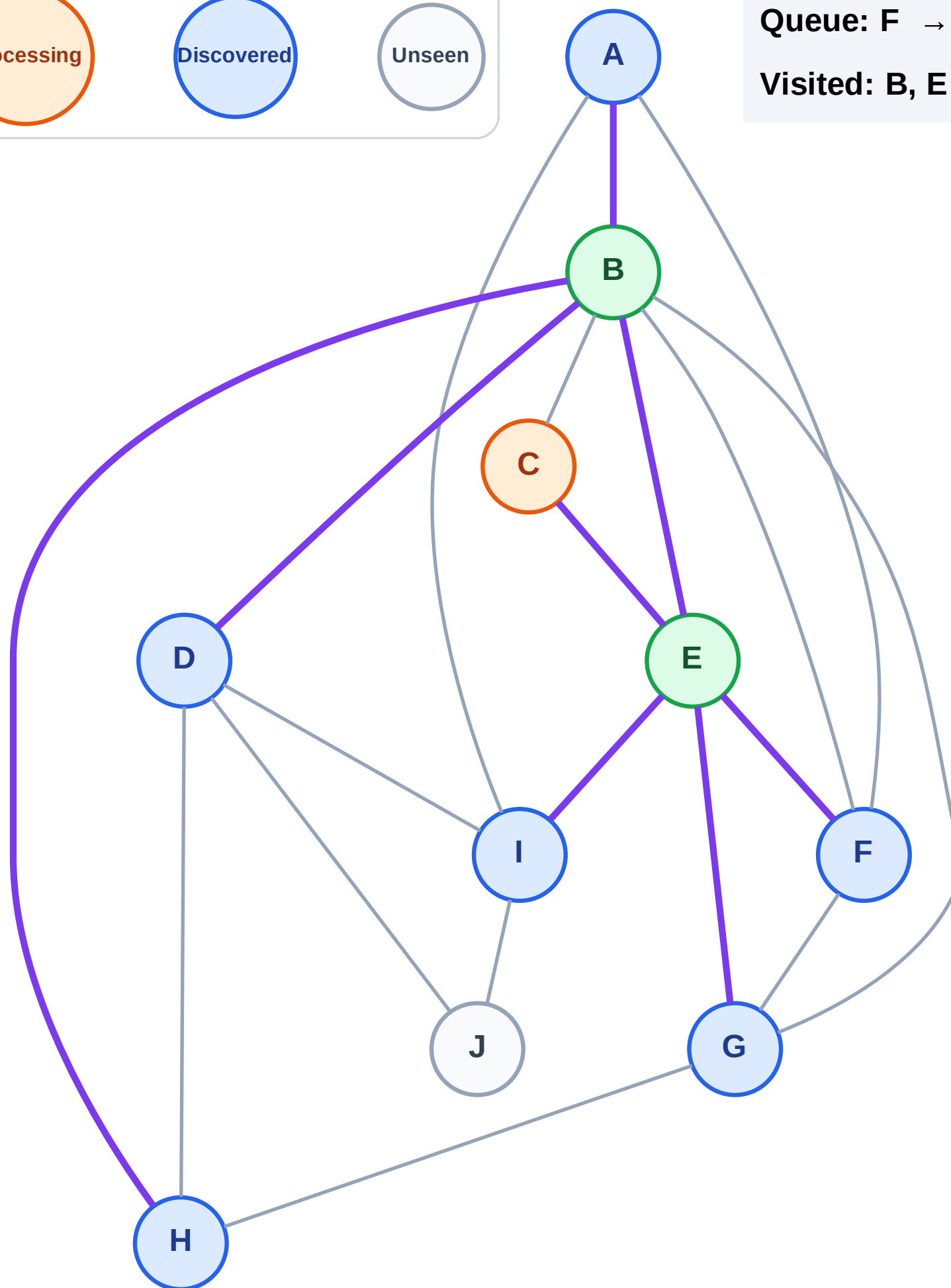
Discovered



Unseen

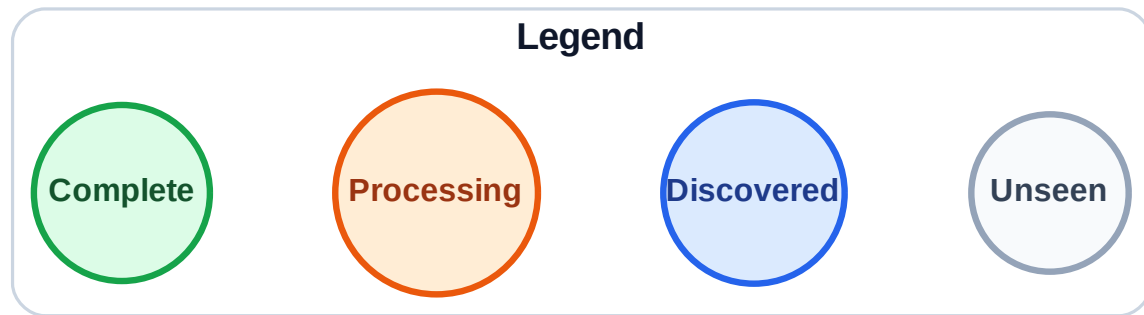
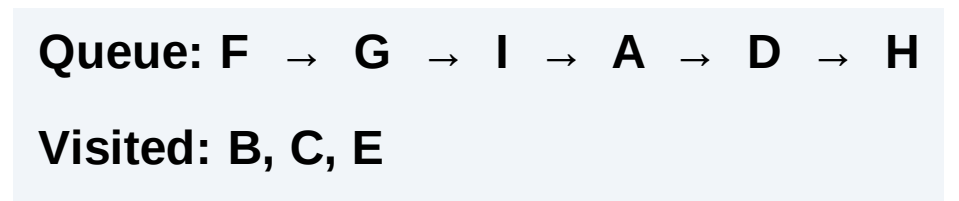
Queue: F → G → I → A → D → H

Visited: B, E



Step 7: No new neighbors from C

Step 7: No new neighbors from C



BFS Traversal

Step 8: Process node F

Legend

Complete

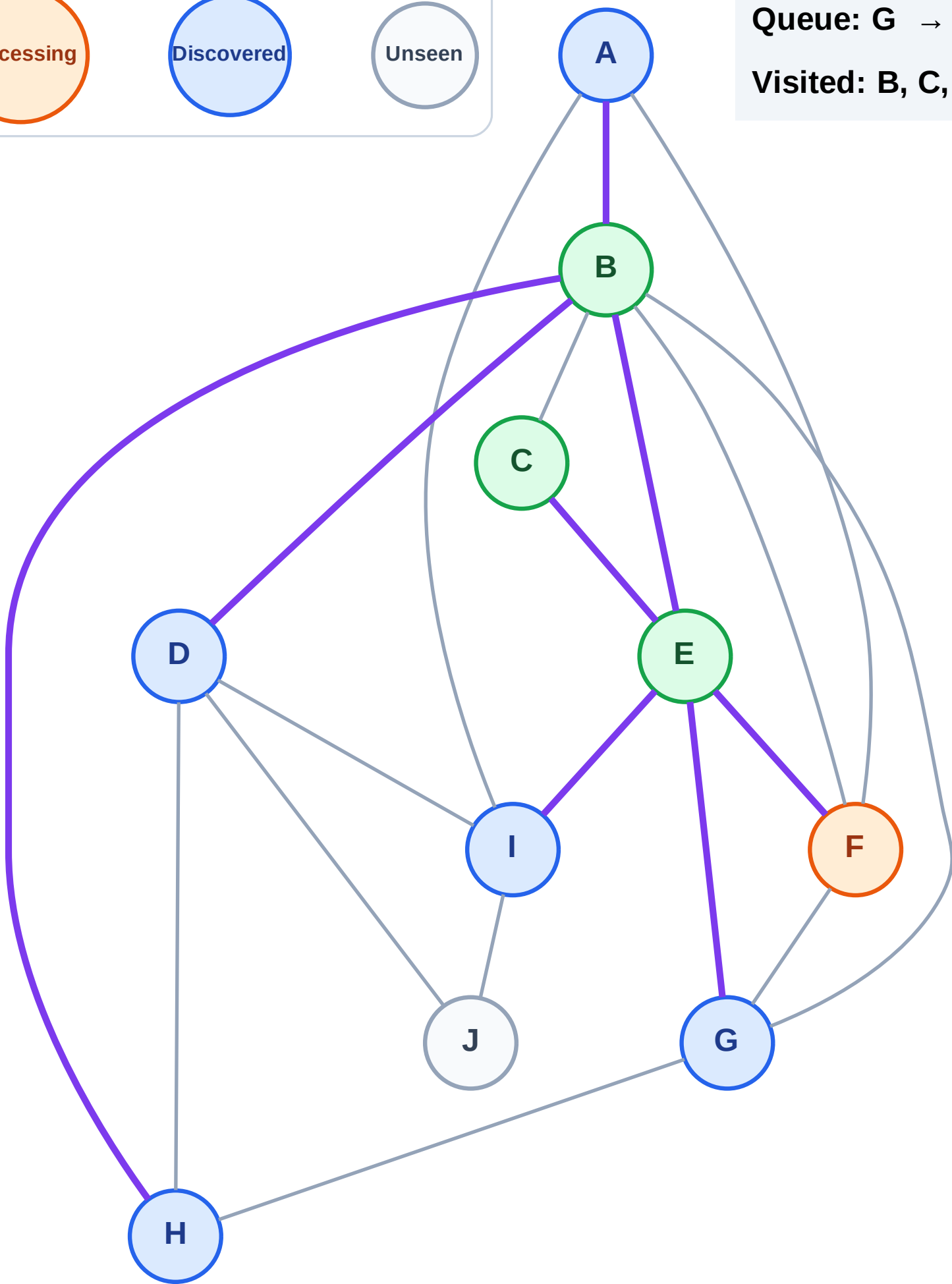
Processing

Discovered

Unseen

Queue: G → I → A → D → H

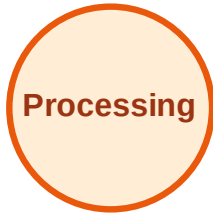
Visited: B, C, E



BFS Traversal

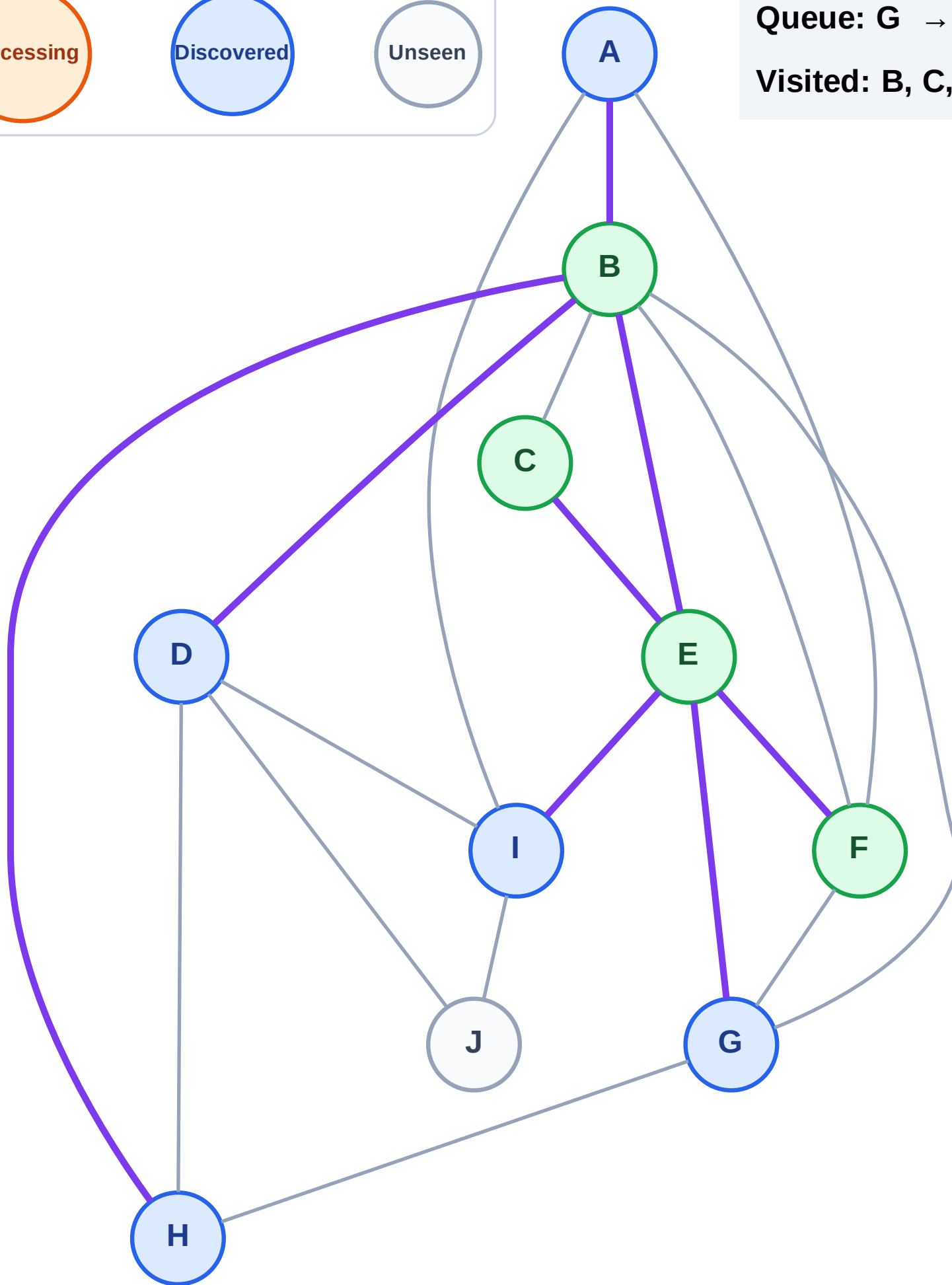
Step 9: No new neighbors from F

Legend



Queue: G → I → A → D → H

Visited: B, C, E, F



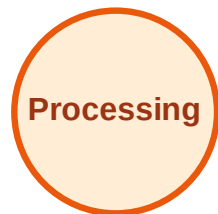
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing

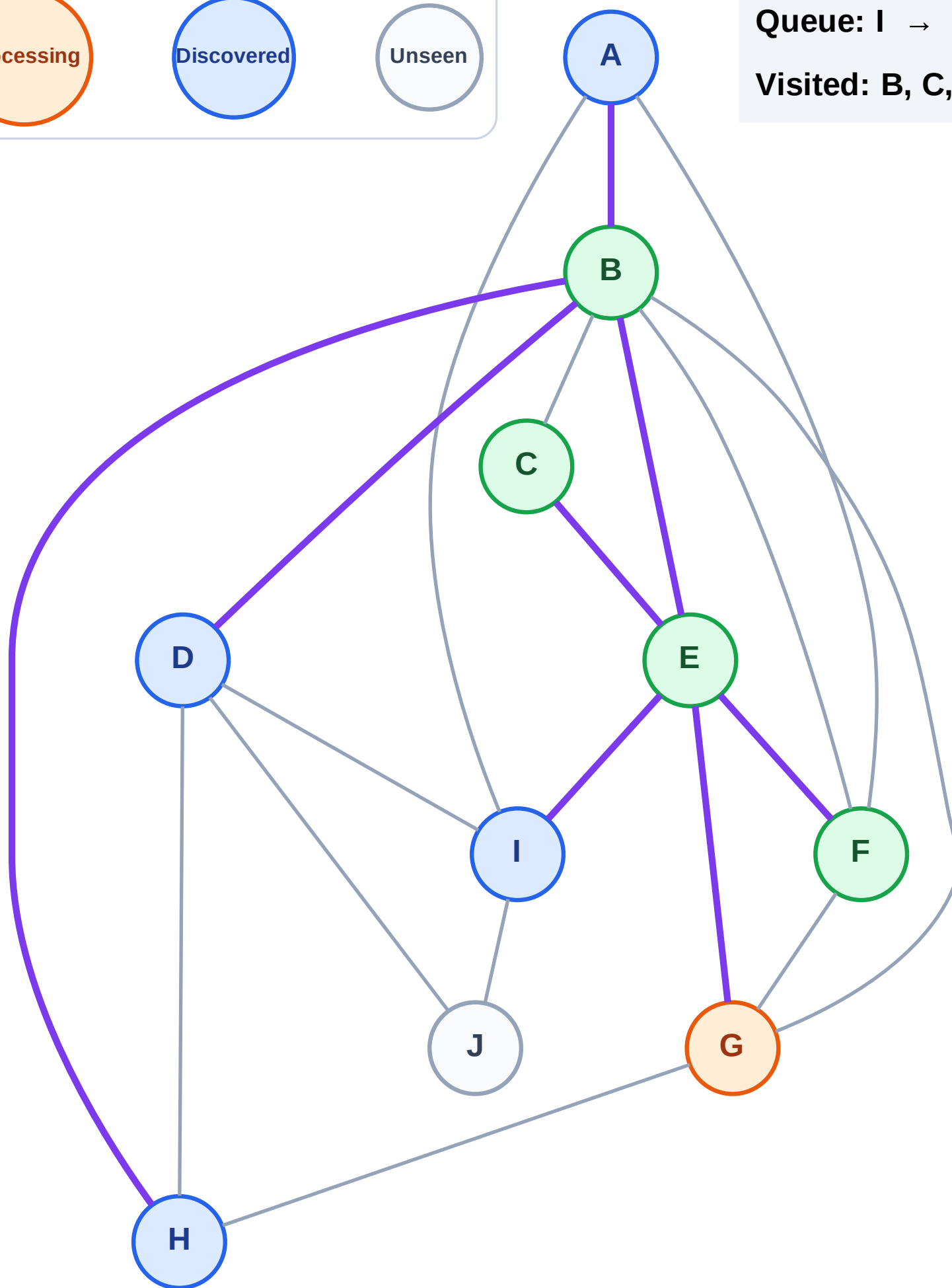


Discovered



Unseen

Queue: I → A → D → H
Visited: B, C, E, F



BFS Traversal

Step 11: No new neighbors from G

Legend

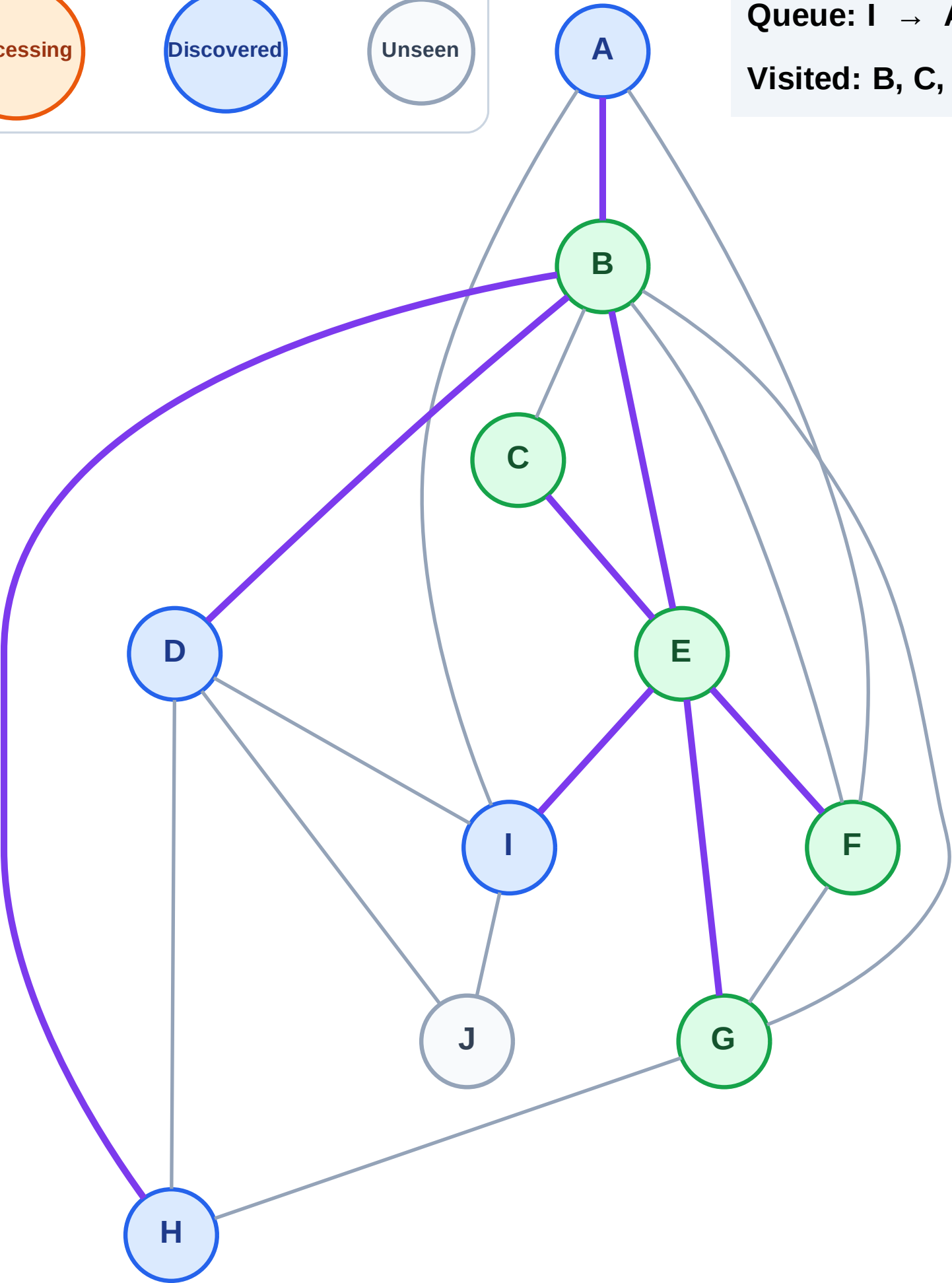
Complete

Processing

Discovered

Unseen

Queue: I → A → D → H
Visited: B, C, E, F, G



BFS Traversal

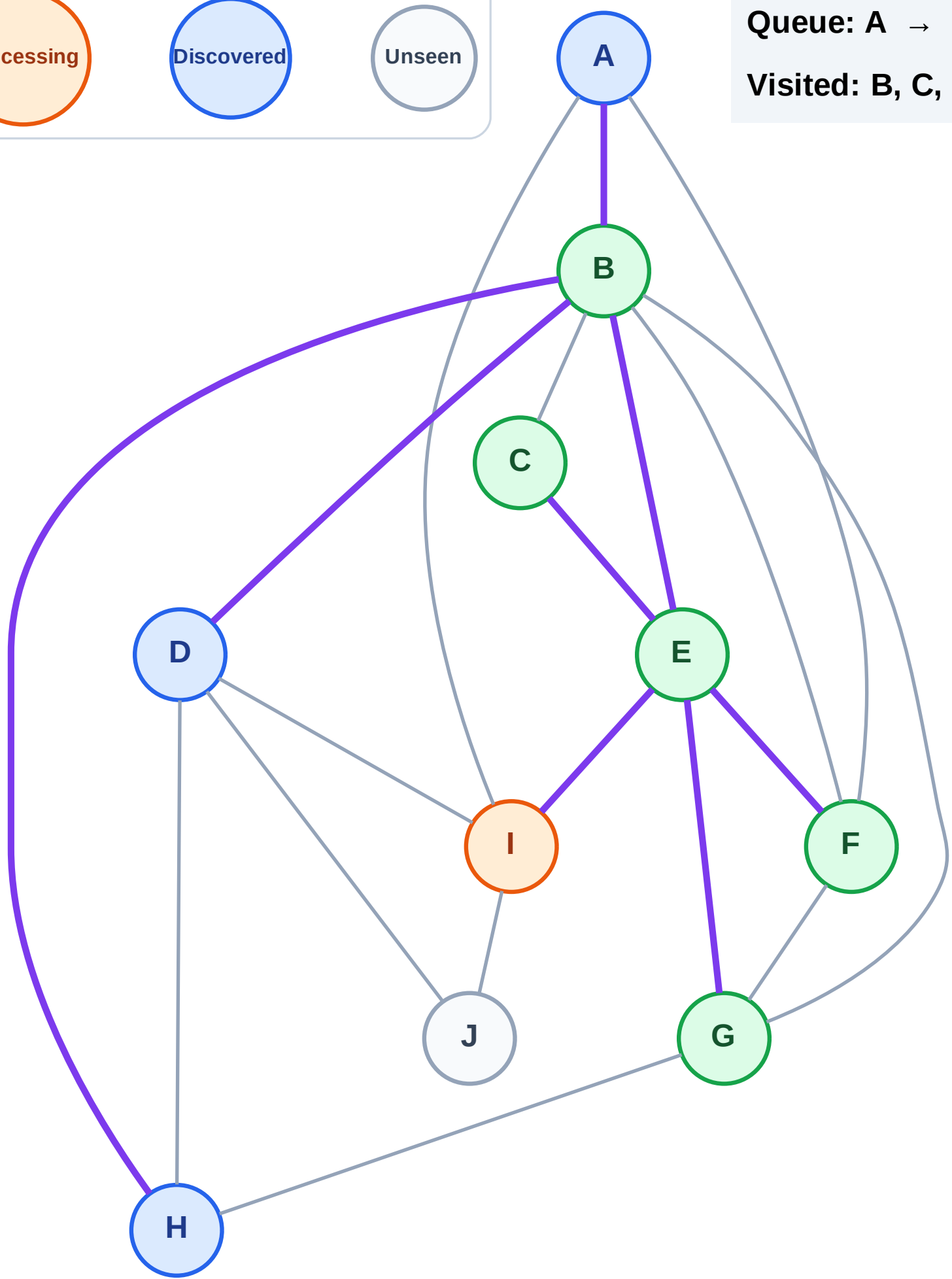
Step 12: Process node I

Legend



Queue: A → D → H

Visited: B, C, E, F, G



BFS Traversal

Step 13: Discover J from I

Legend

Complete

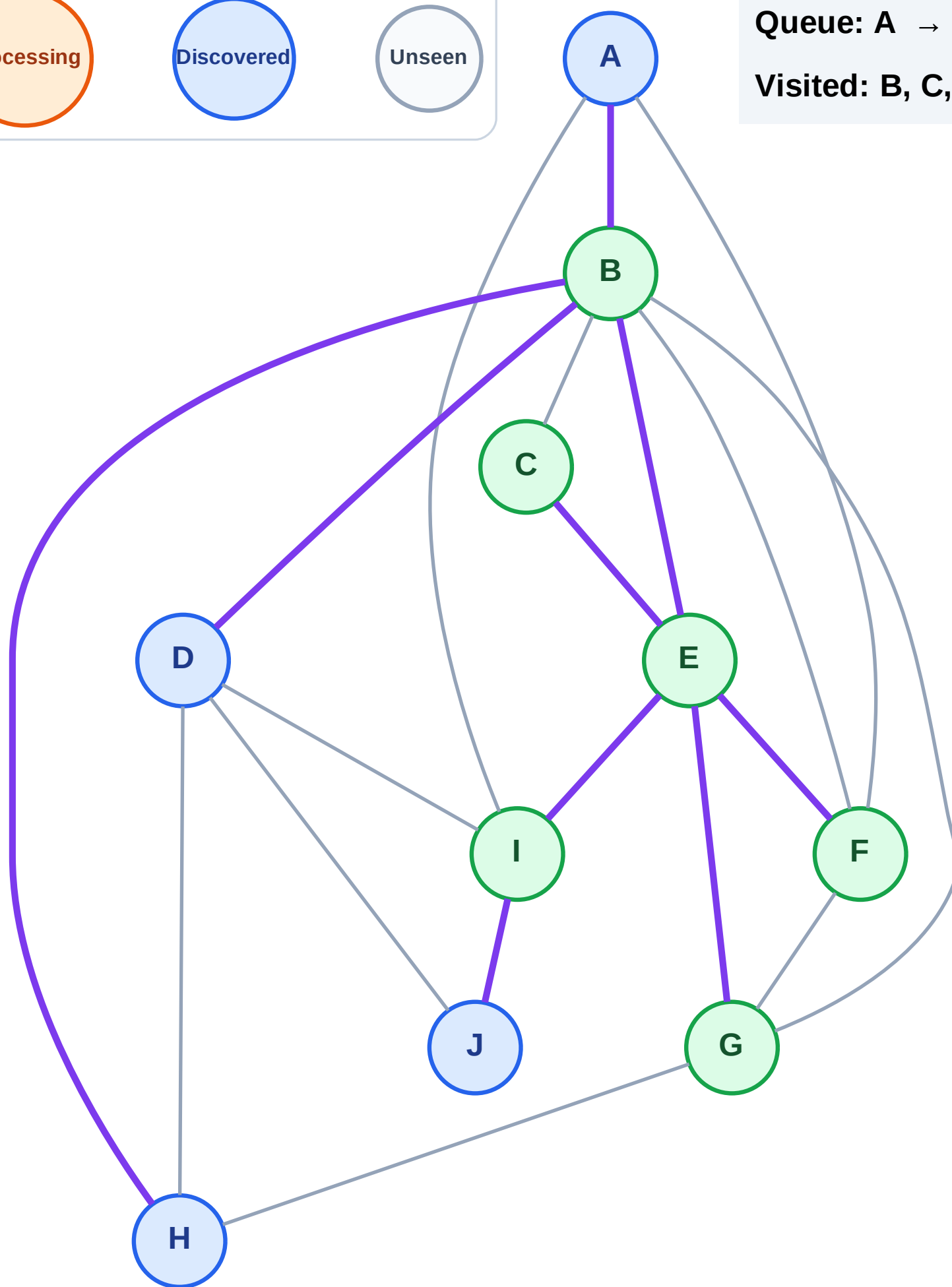
Processing

Discovered

Unseen

Queue: A → D → H → J

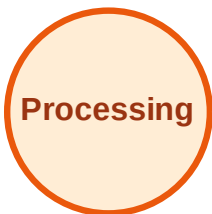
Visited: B, C, E, F, G, I



BFS Traversal

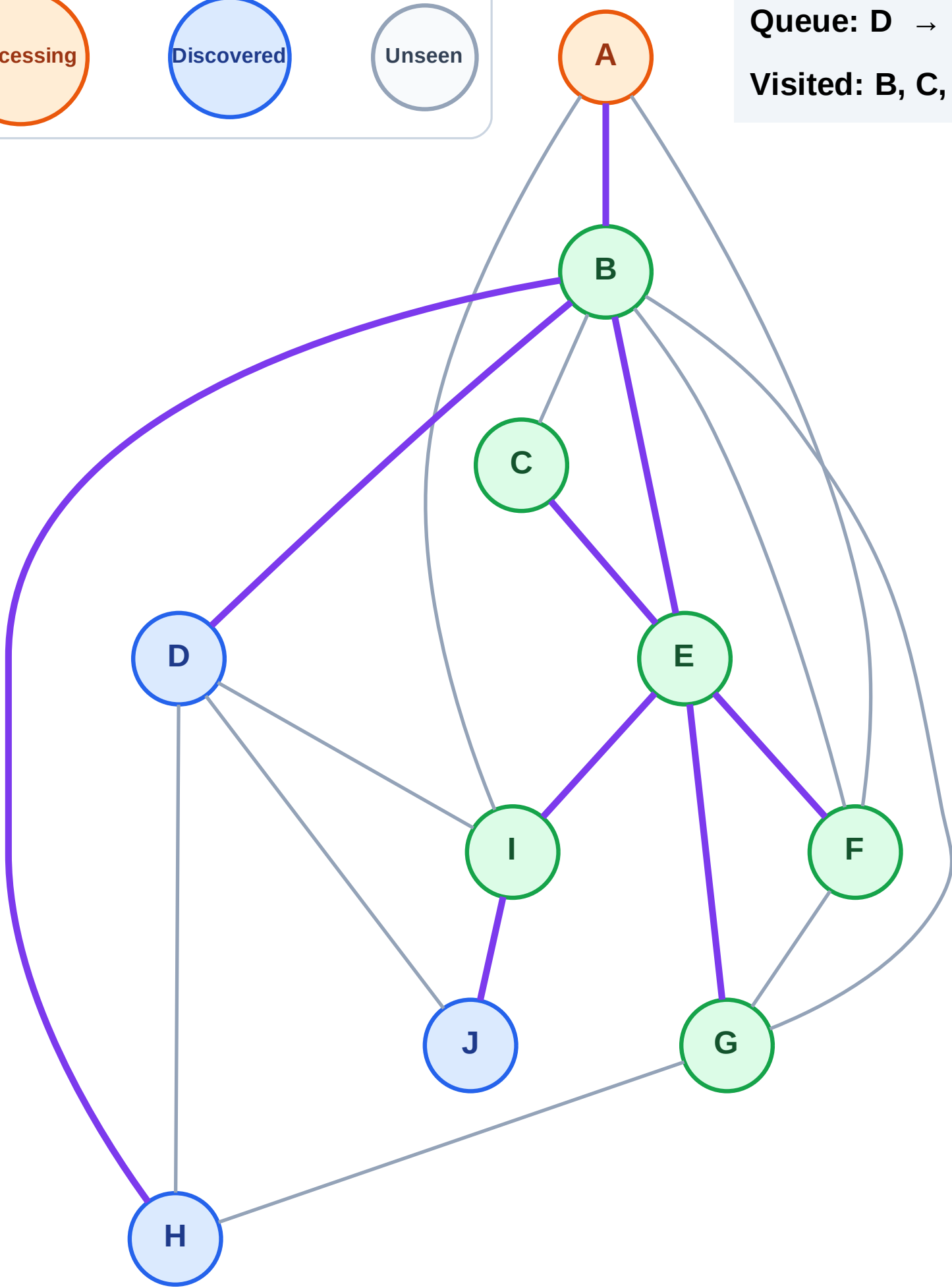
Step 14: Process node A

Legend



Queue: D → H → J

Visited: B, C, E, F, G, I



BFS Traversal

Step 15: No new neighbors from A

Legend

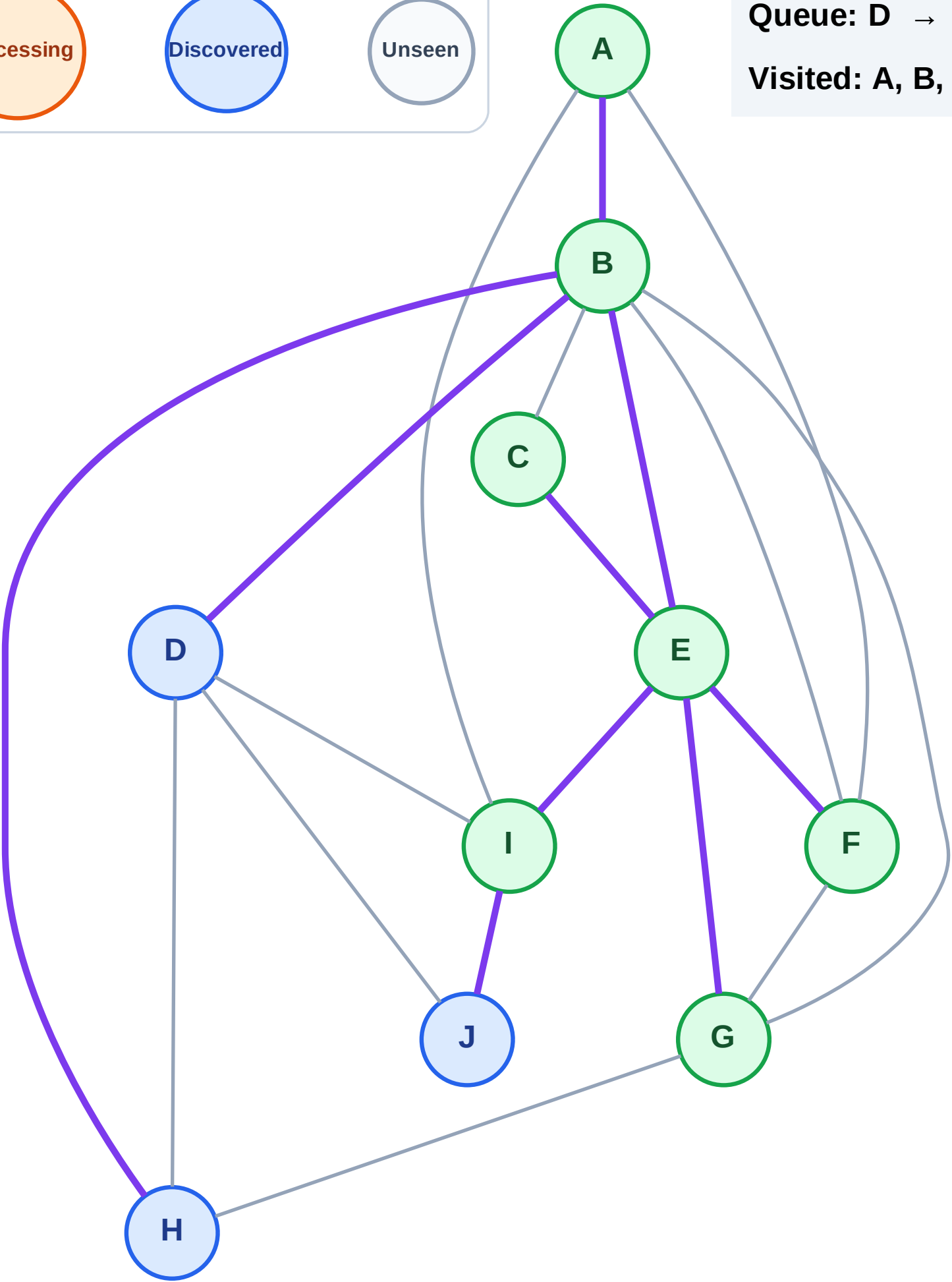
Complete

Processing

Discovered

Unseen

Queue: D → H → J
Visited: A, B, C, E, F, G, I



BFS Traversal

Step 16: Process node D

Legend

Complete

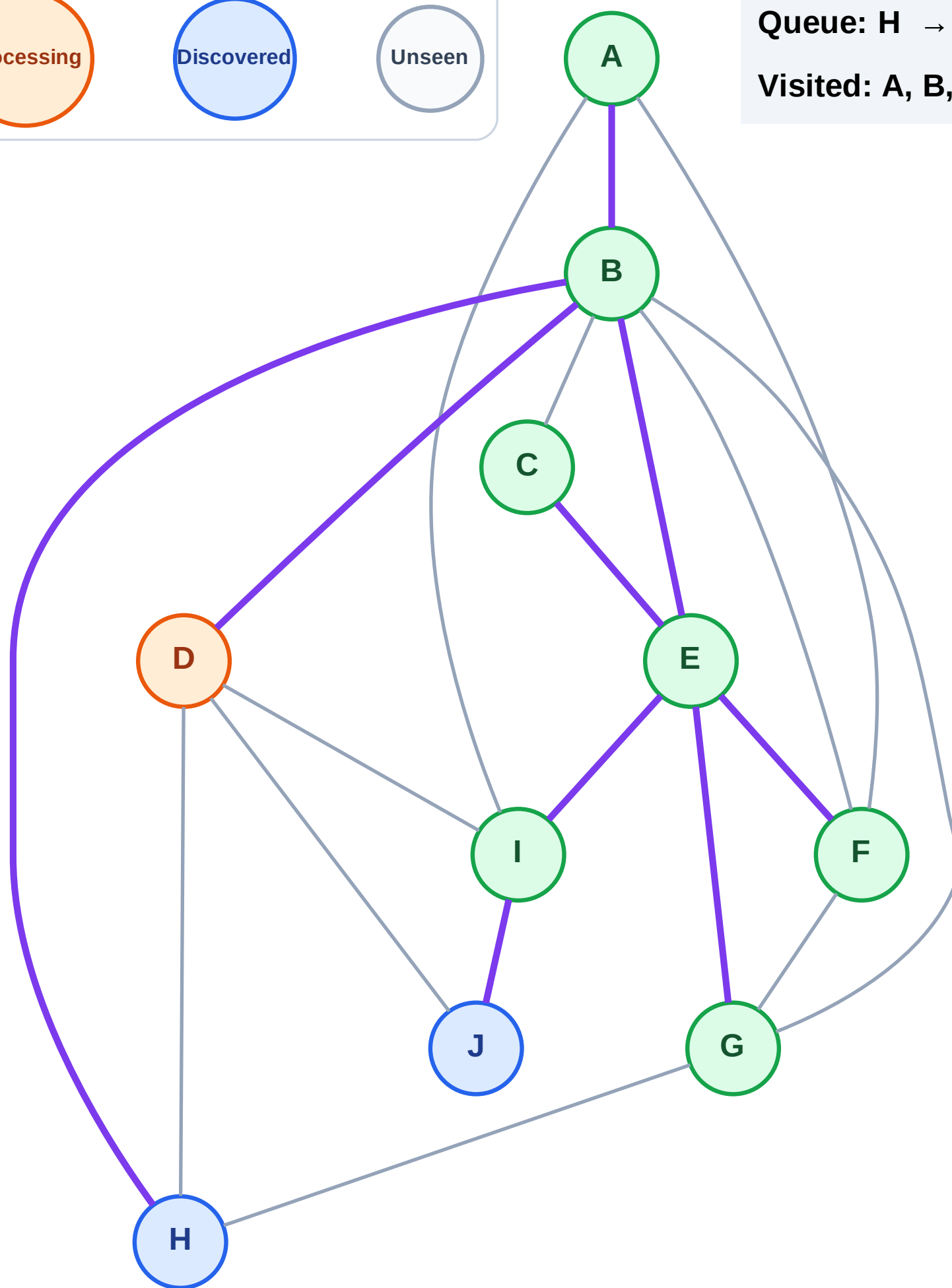
Processing

Discovered

Unseen

Queue: H → J

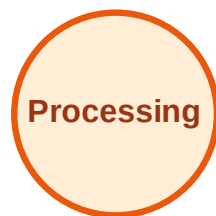
Visited: A, B, C, E, F, G, I



BFS Traversal

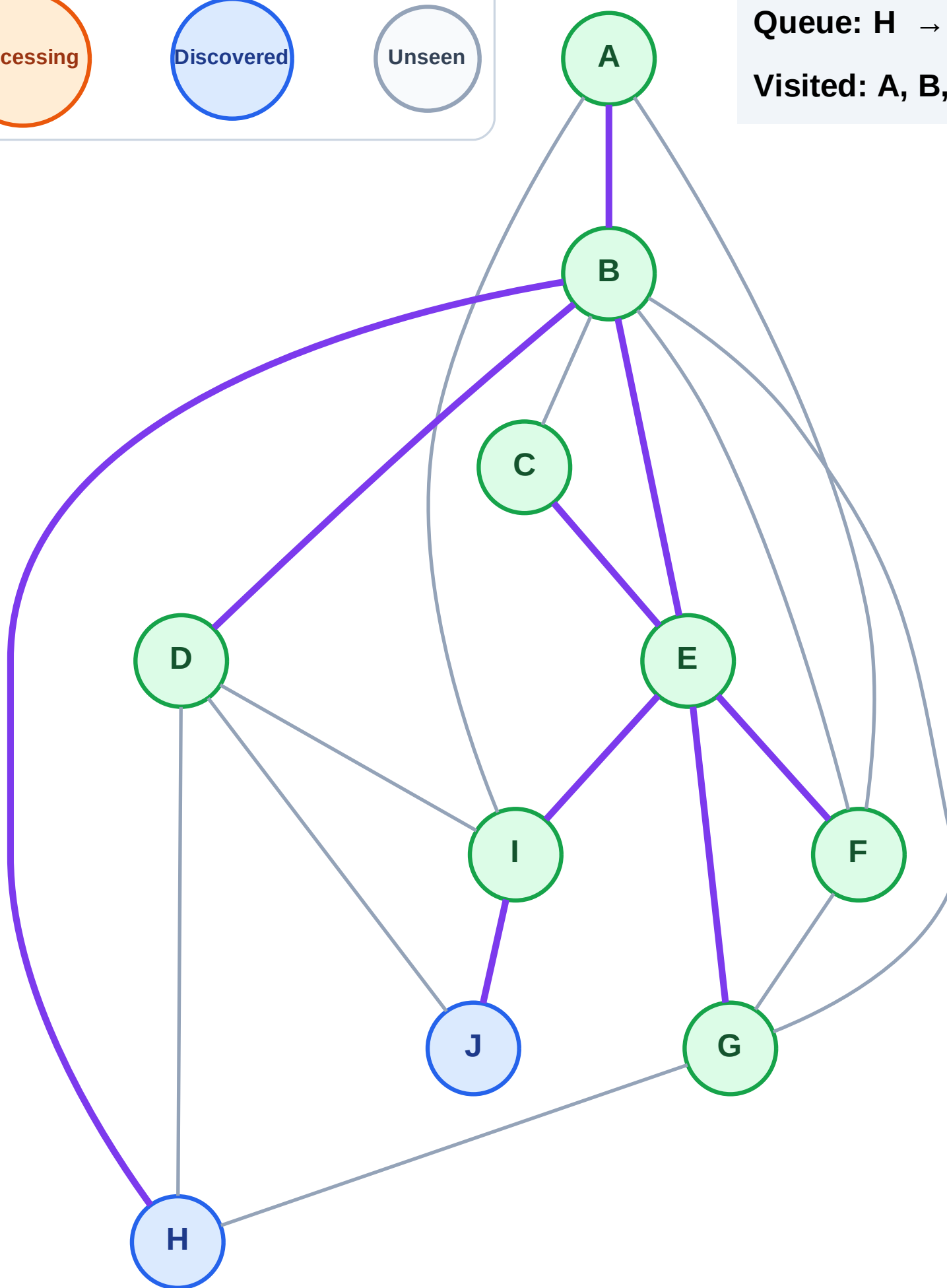
Step 17: No new neighbors from D

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node H

Legend

Complete

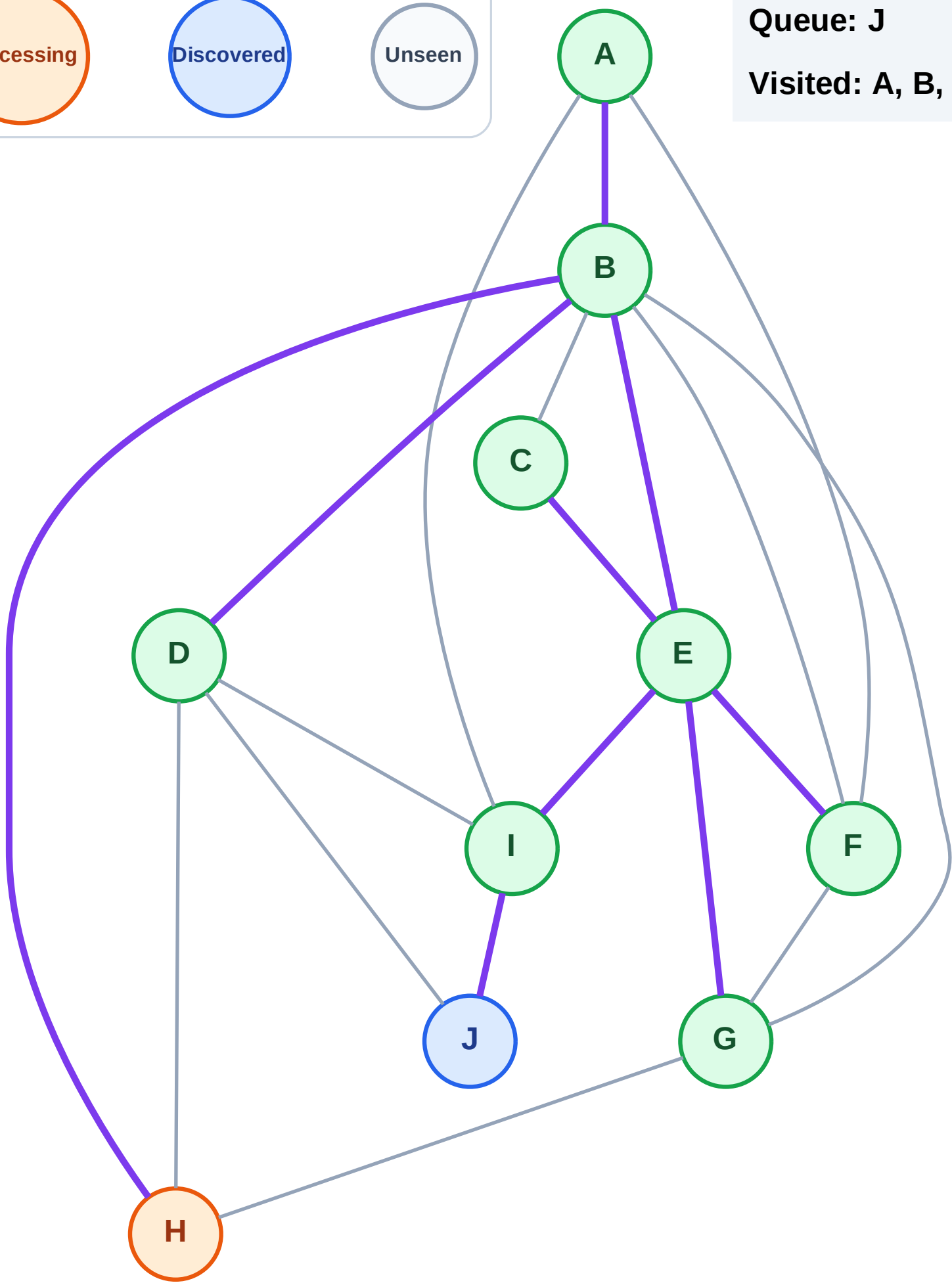
Processing

Discovered

Unseen

Queue: J

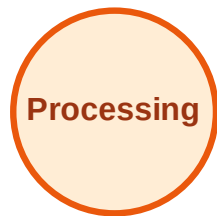
Visited: A, B, C, D, E, F, G, I



BFS Traversal

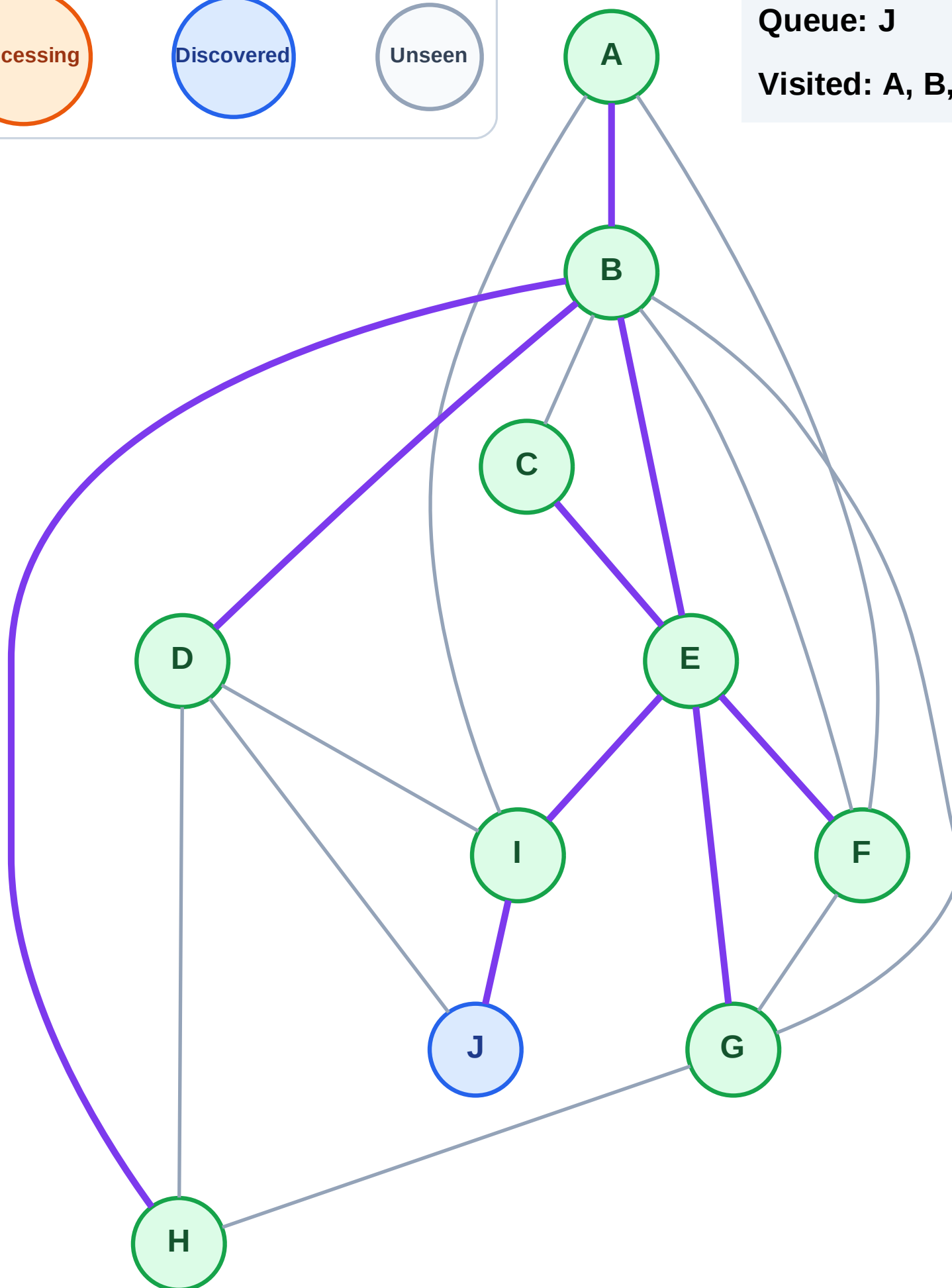
Step 19: No new neighbors from H

Legend



Queue: J

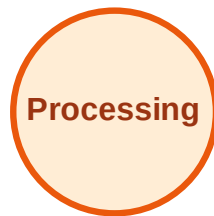
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

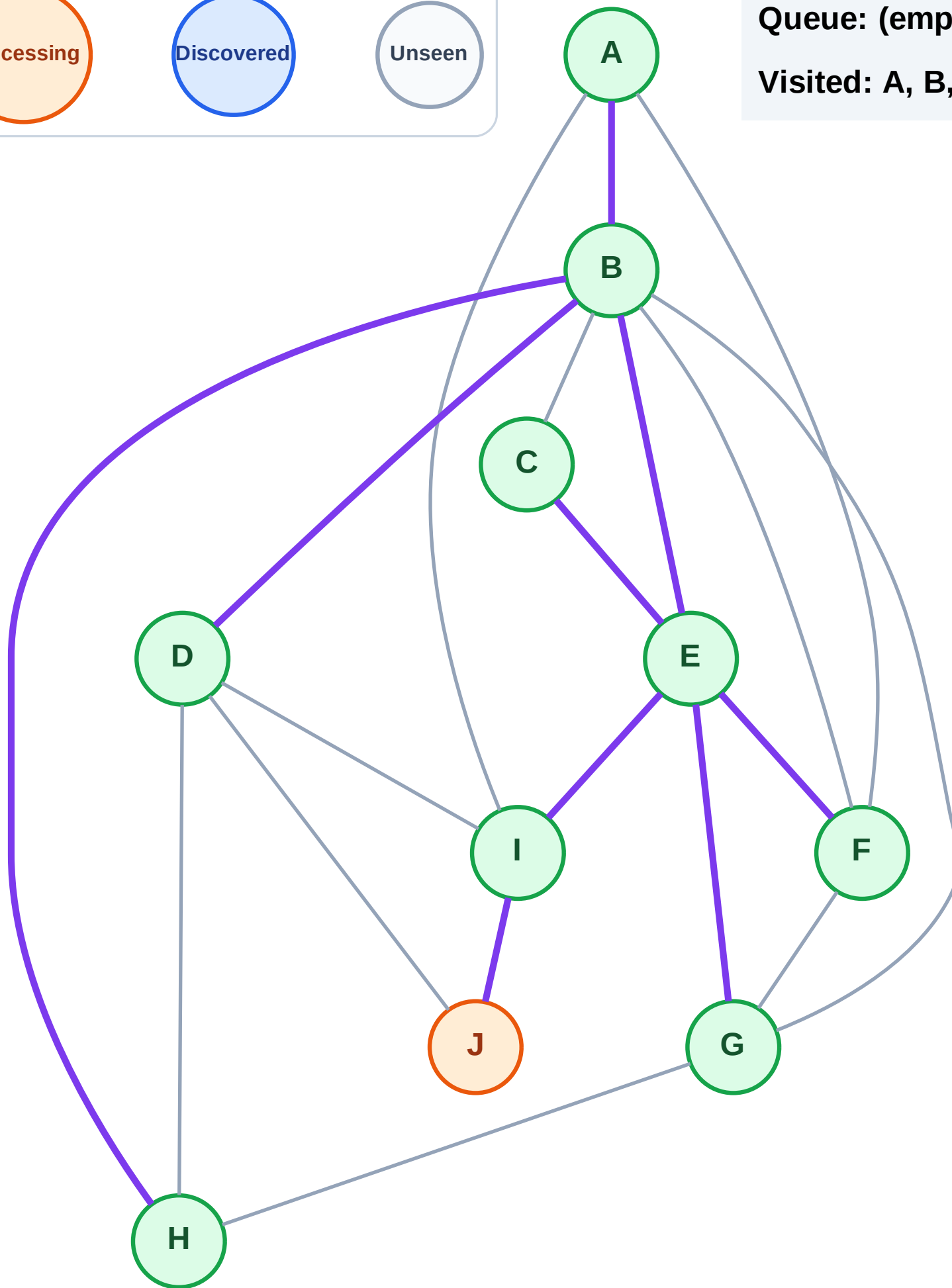
Step 20: Process node J

Legend



Queue: (empty)

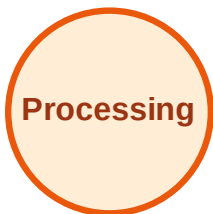
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

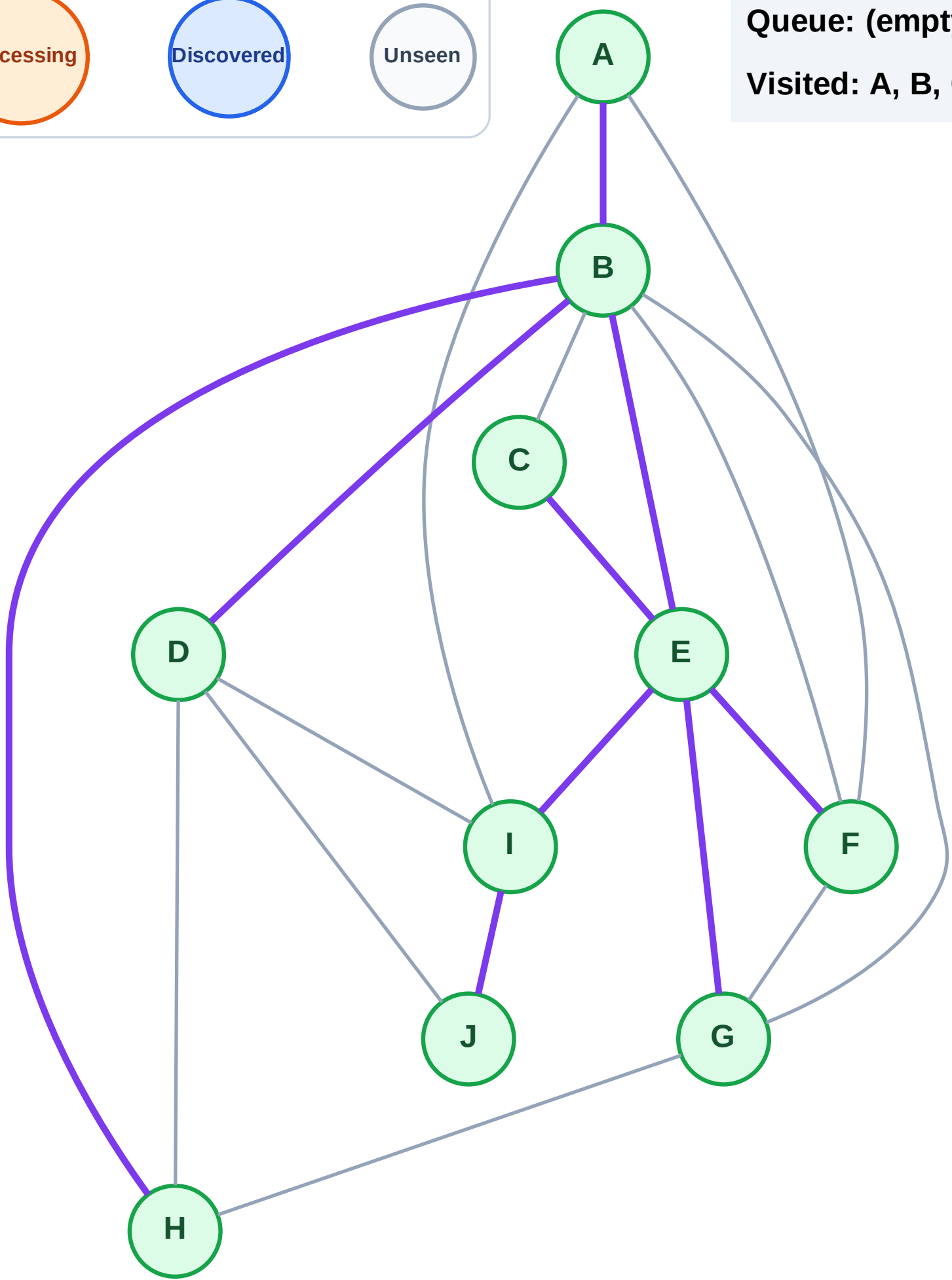
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

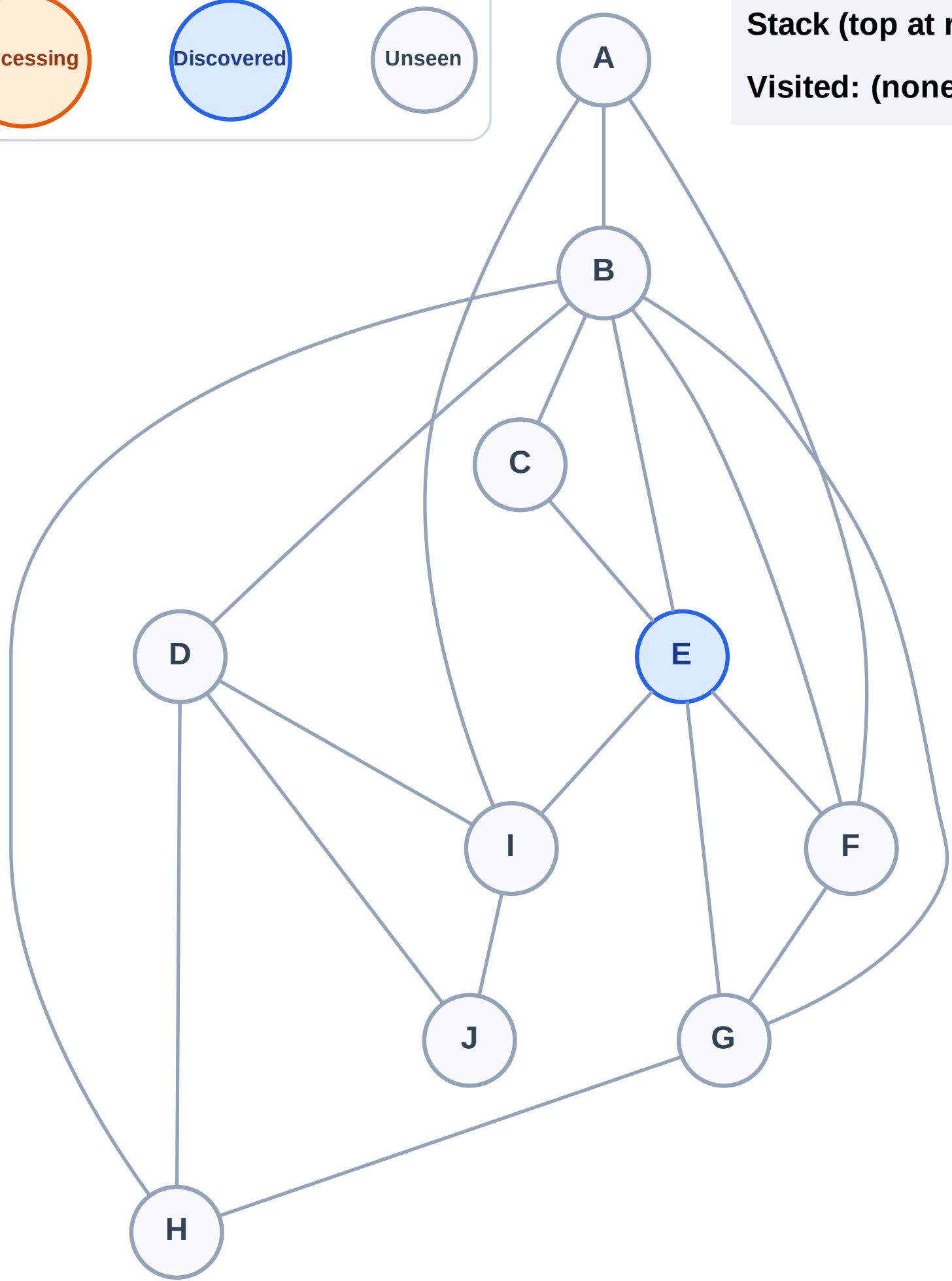
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

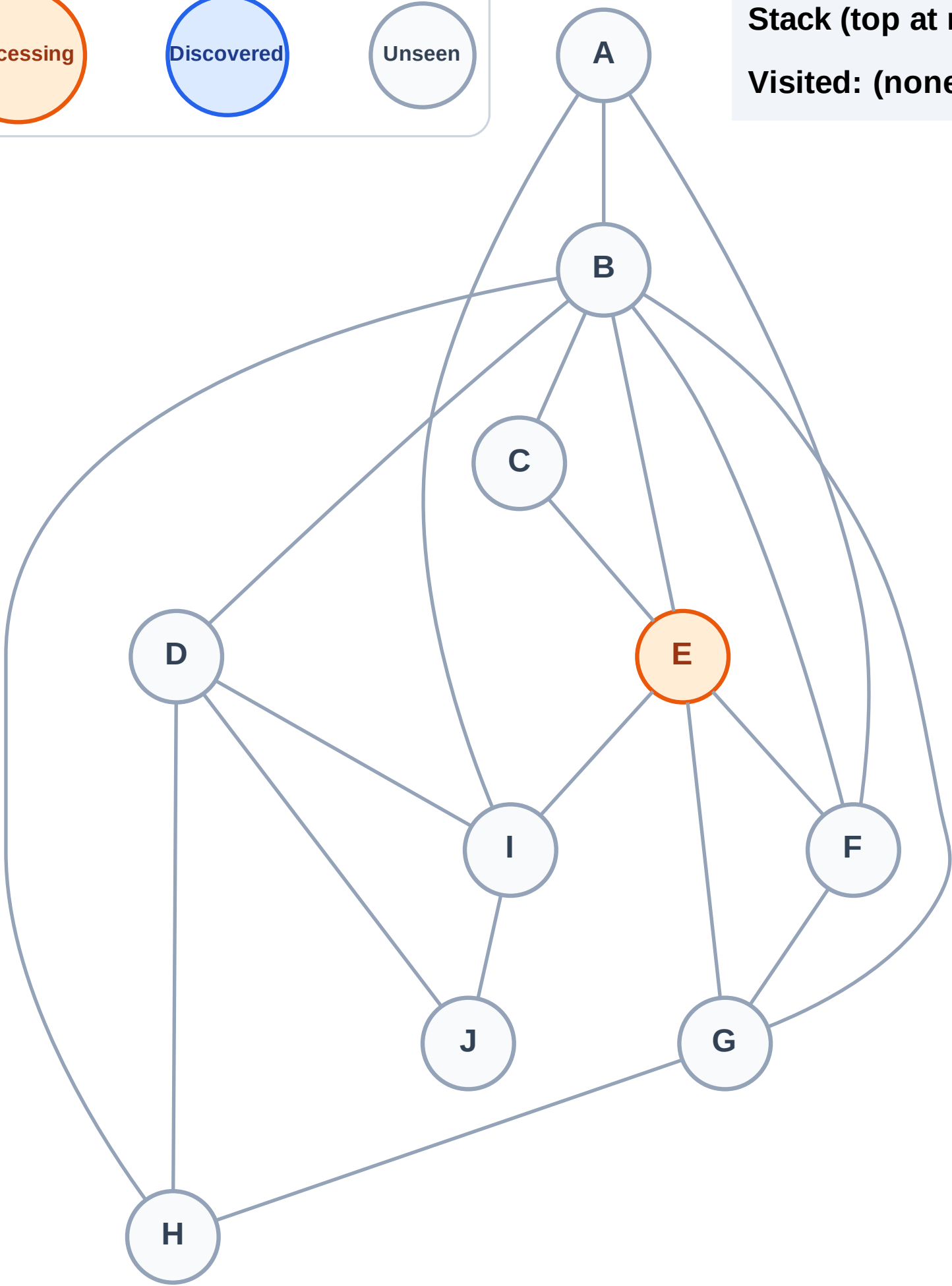
Complete

Processing

Discovered

Unseen

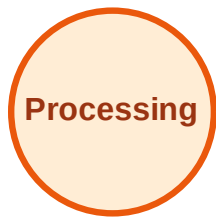
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

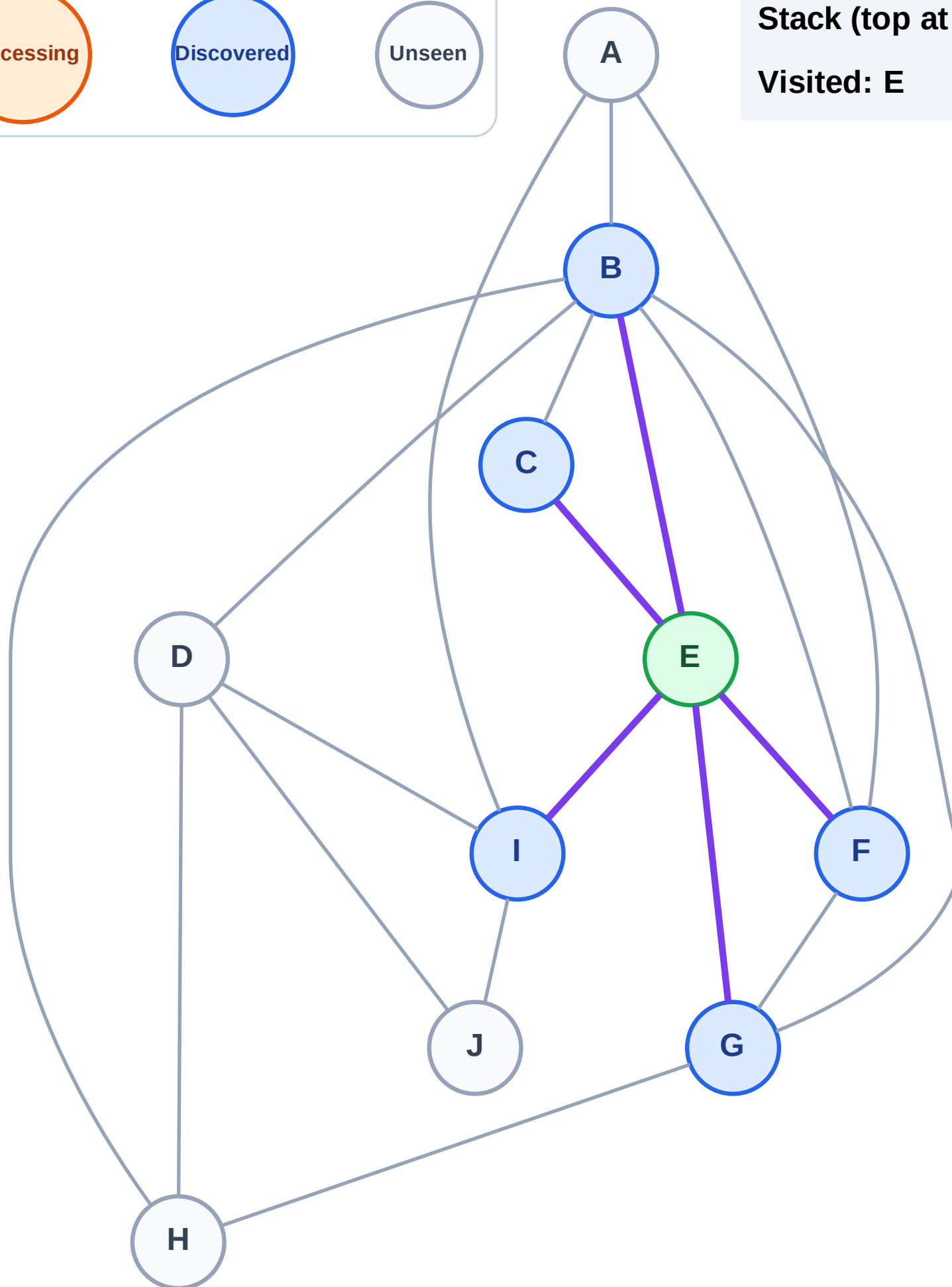
Step 3: Discover I, G, F, C, B from E

Legend



Stack (top at right): I | G | F | C | B

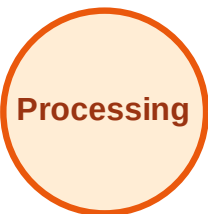
Visited: E



DFS Traversal

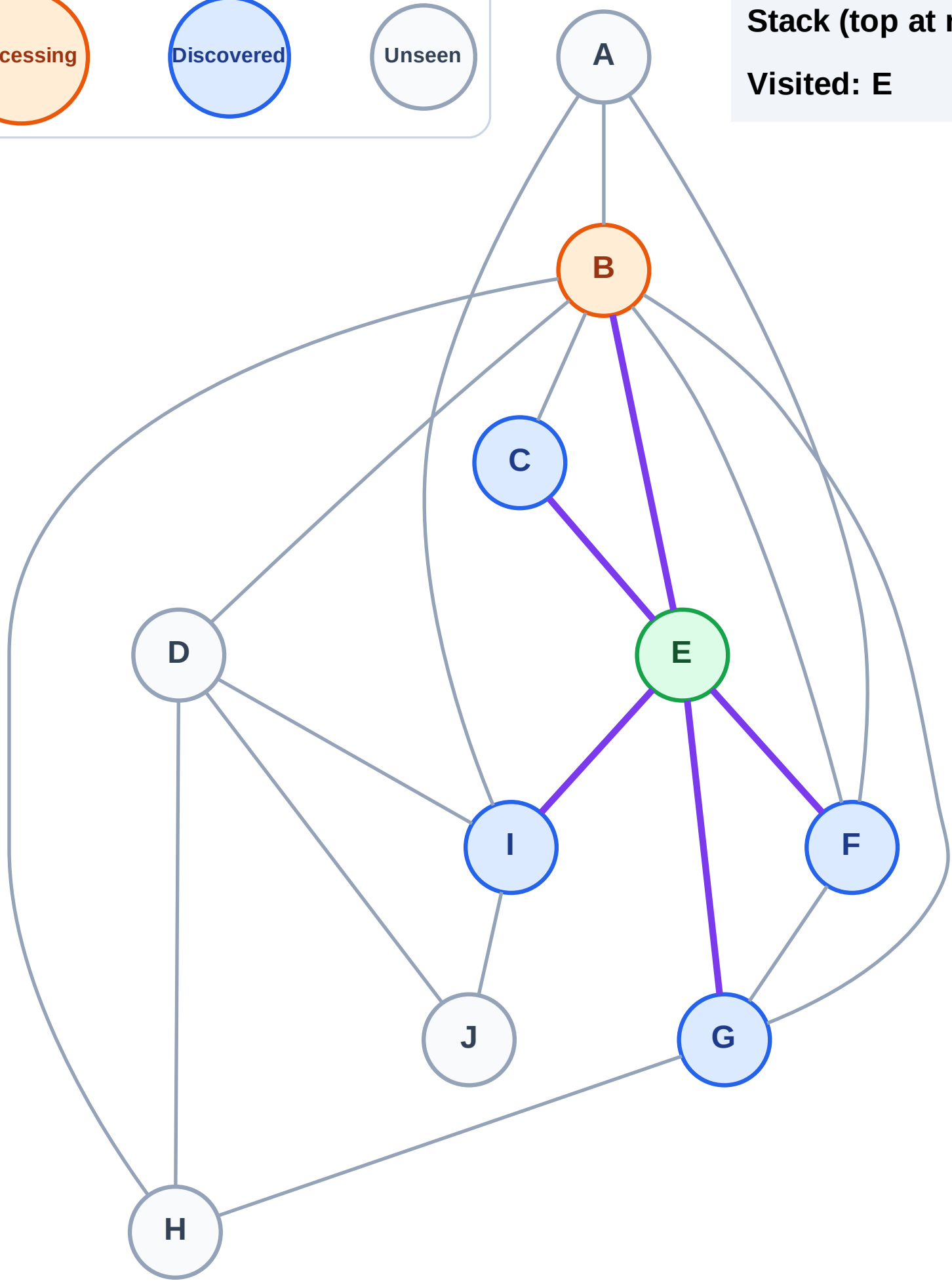
Step 4: Process node B

Legend



Stack (top at right): I | G | F | C

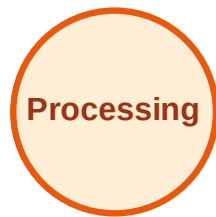
Visited: E



DFS Traversal

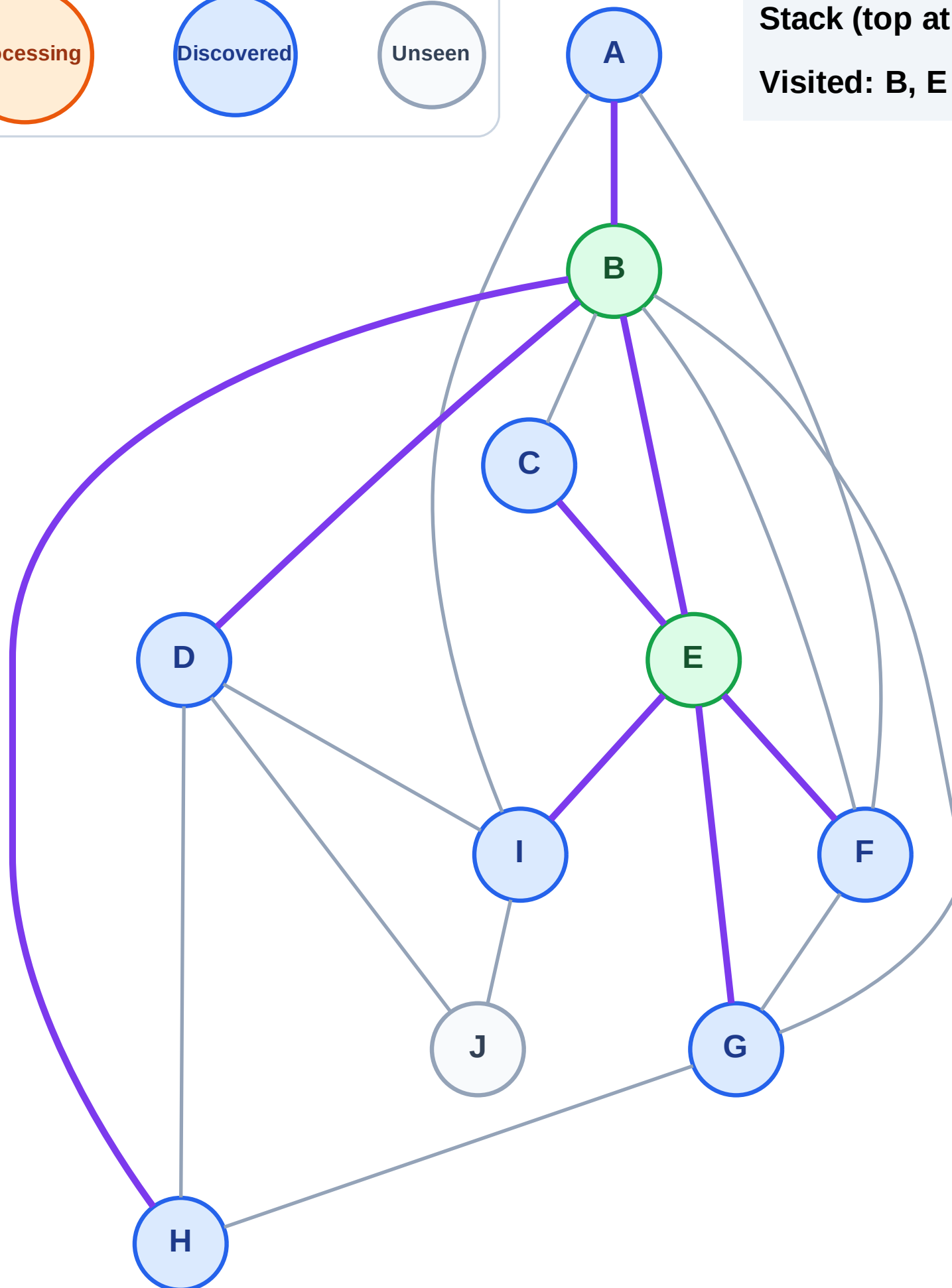
Step 5: Discover H, D, A from B

Legend



Stack (top at right): I | G | F | C | H | D | A

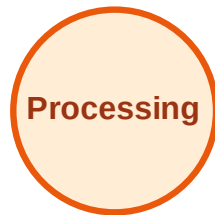
Visited: B, E



DFS Traversal

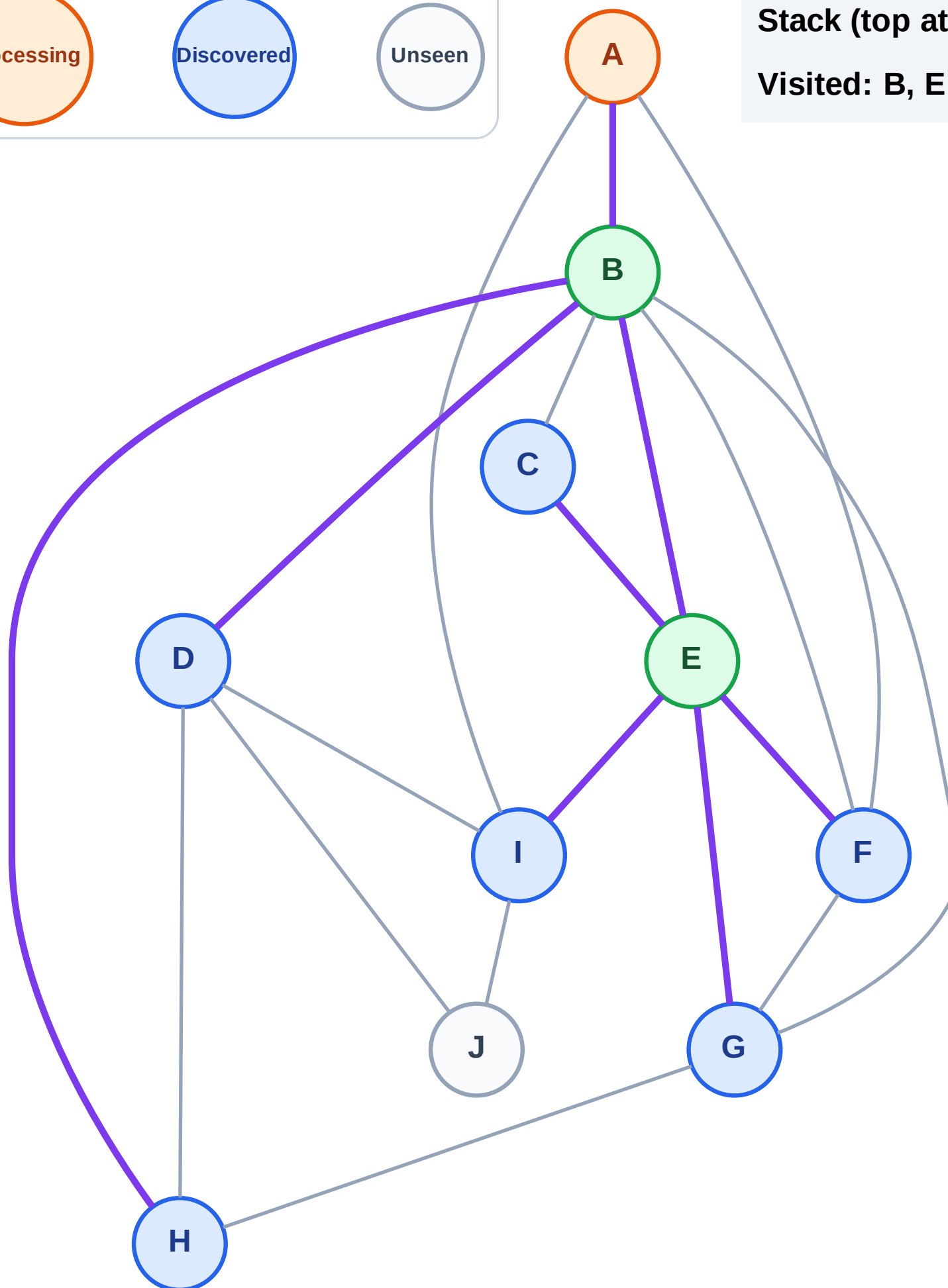
Step 6: Process node A

Legend



Stack (top at right): I | G | F | C | H | D

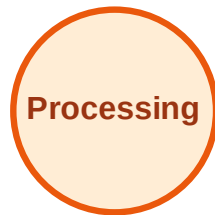
Visited: B, E



DFS Traversal

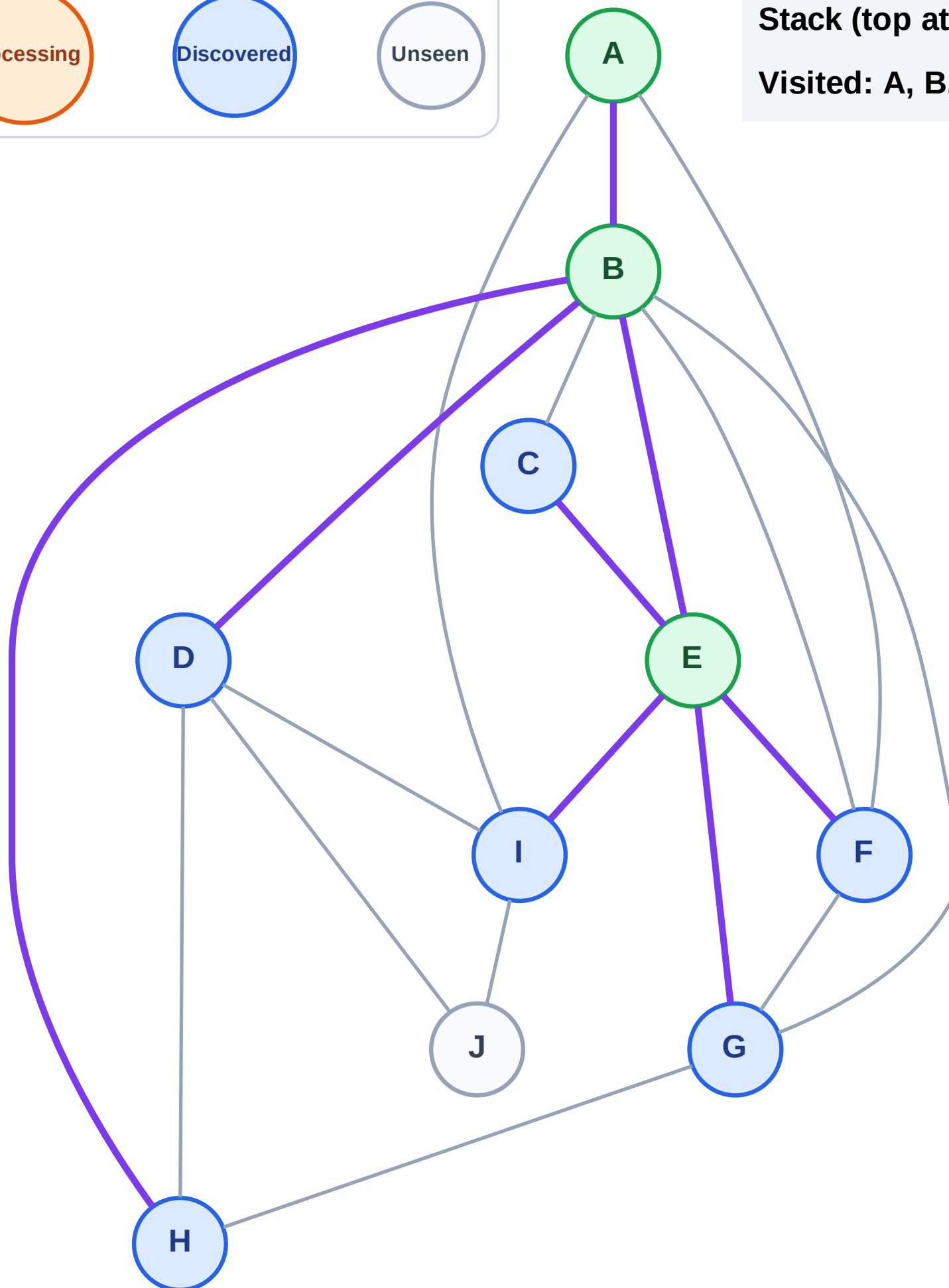
Step 7: No new neighbors from A

Legend

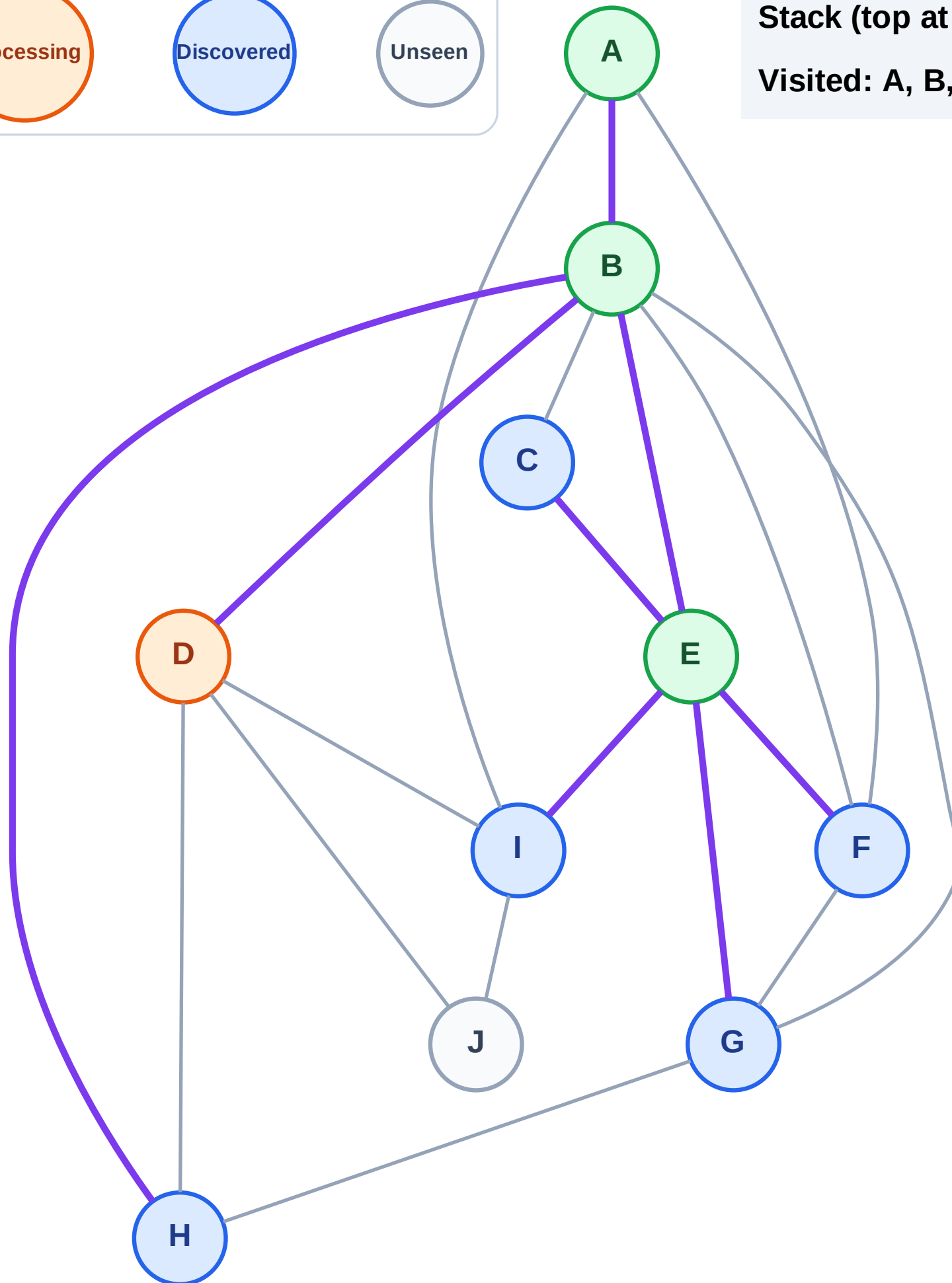
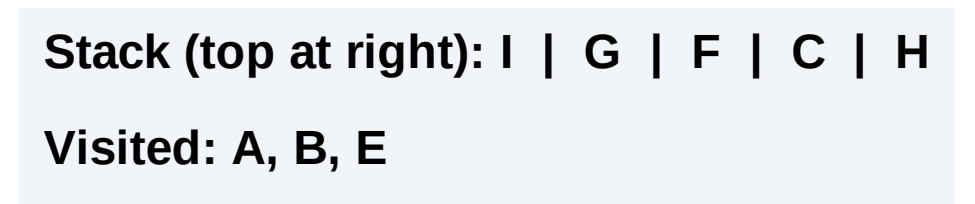


Stack (top at right): I | G | F | C | H | D

Visited: A, B, E



Step 8: Process node D



DFS Traversal

Step 9: Discover J from D

Legend

Complete

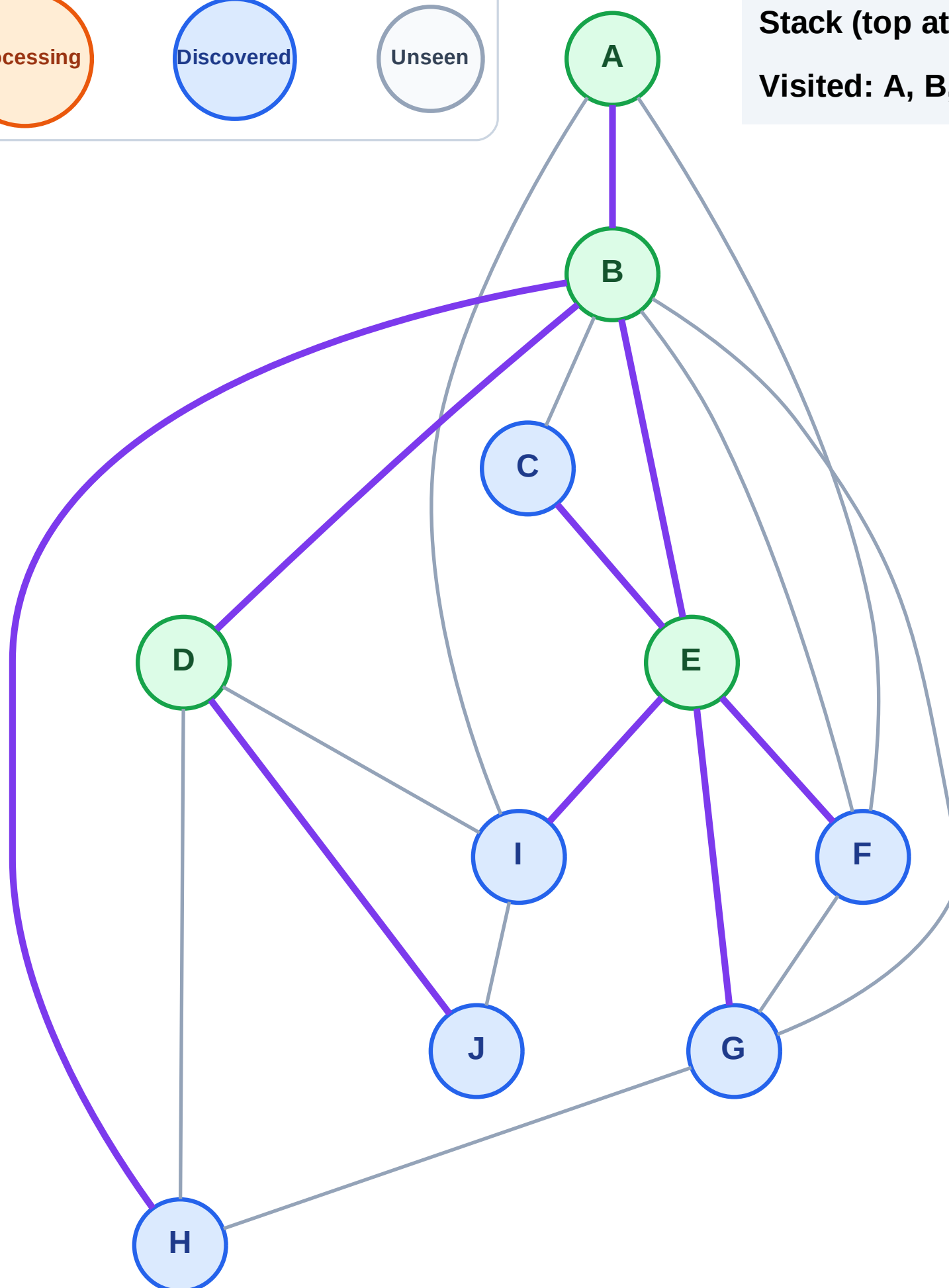
Processing

Discovered

Unseen

Stack (top at right): I | G | F | C | H | J

Visited: A, B, D, E



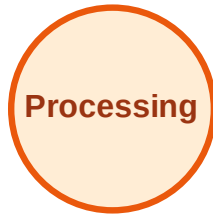
DFS Traversal

Step 10: Process node J

Legend



Complete



Processing



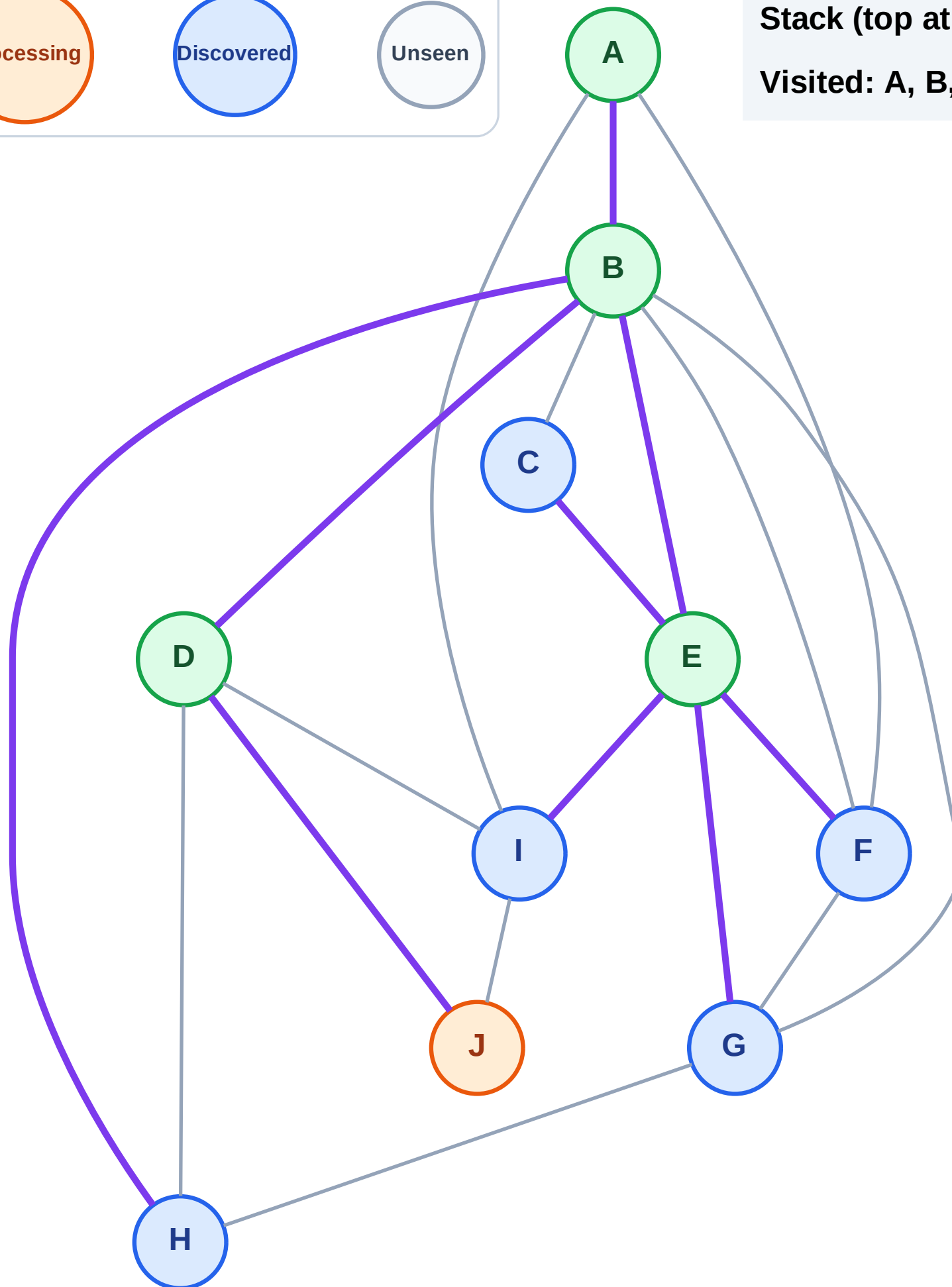
Discovered



Unseen

Stack (top at right): I | G | F | C | H

Visited: A, B, D, E



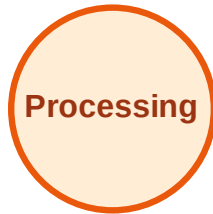
DFS Traversal

Step 11: No new neighbors from J

Legend



Complete



Processing



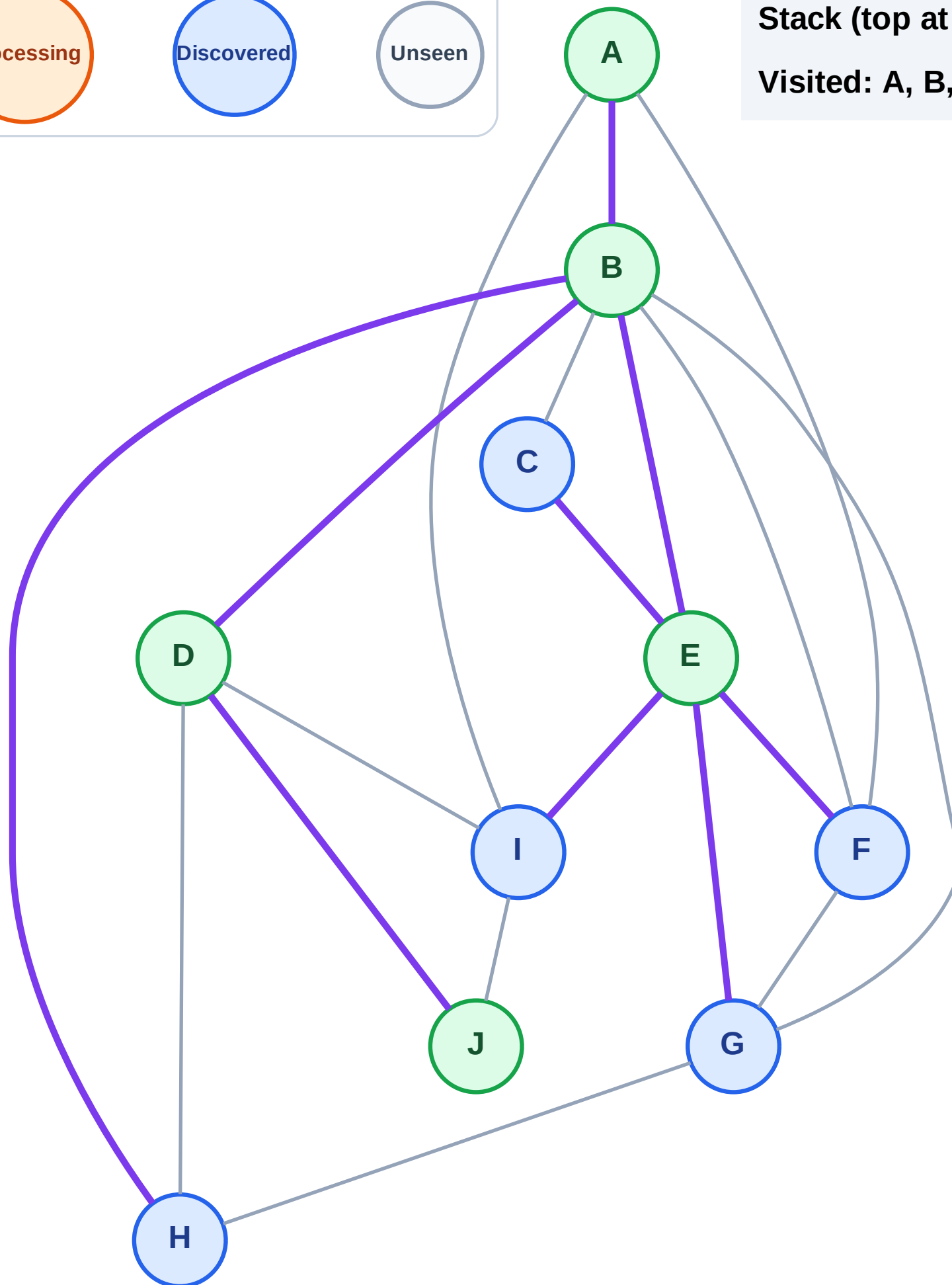
Discovered



Unseen

Stack (top at right): I | G | F | C | H

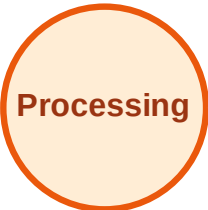
Visited: A, B, D, E, J



DFS Traversal

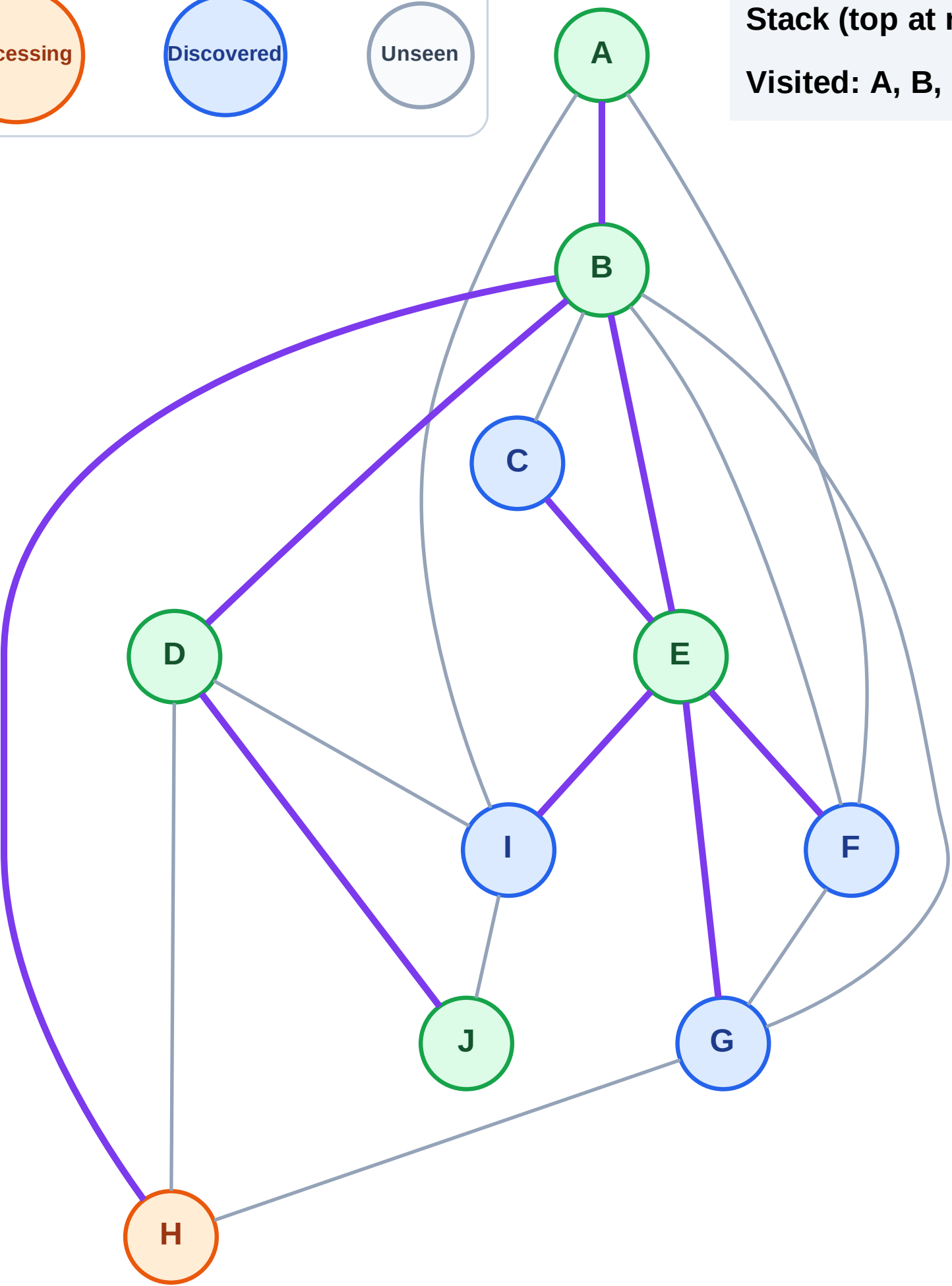
Step 12: Process node H

Legend



Stack (top at right): I | G | F | C

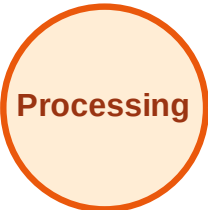
Visited: A, B, D, E, J



DFS Traversal

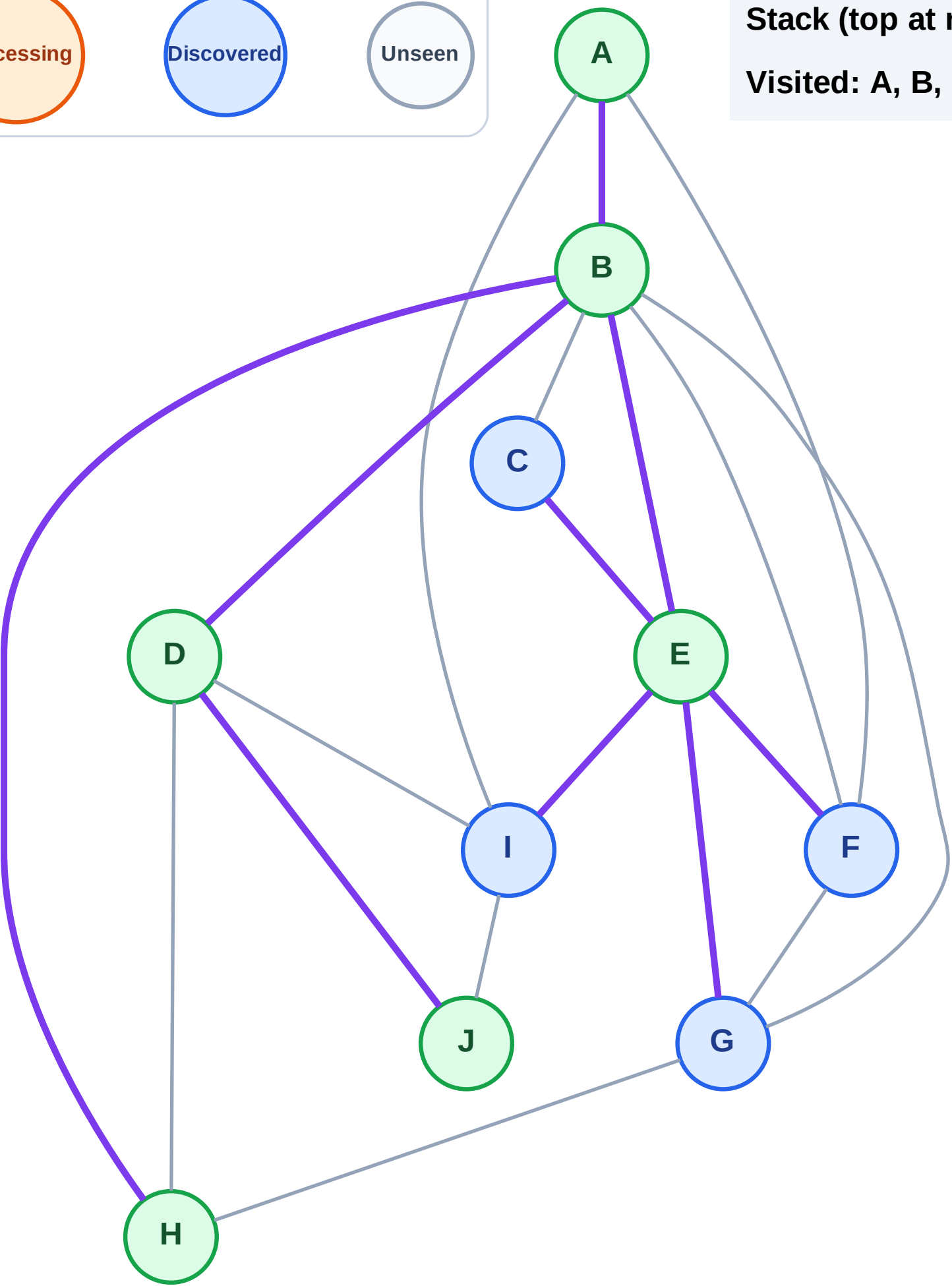
Step 13: No new neighbors from H

Legend



Stack (top at right): I | G | F | C

Visited: A, B, D, E, H, J



DFS Traversal

Step 14: Process node C

Legend

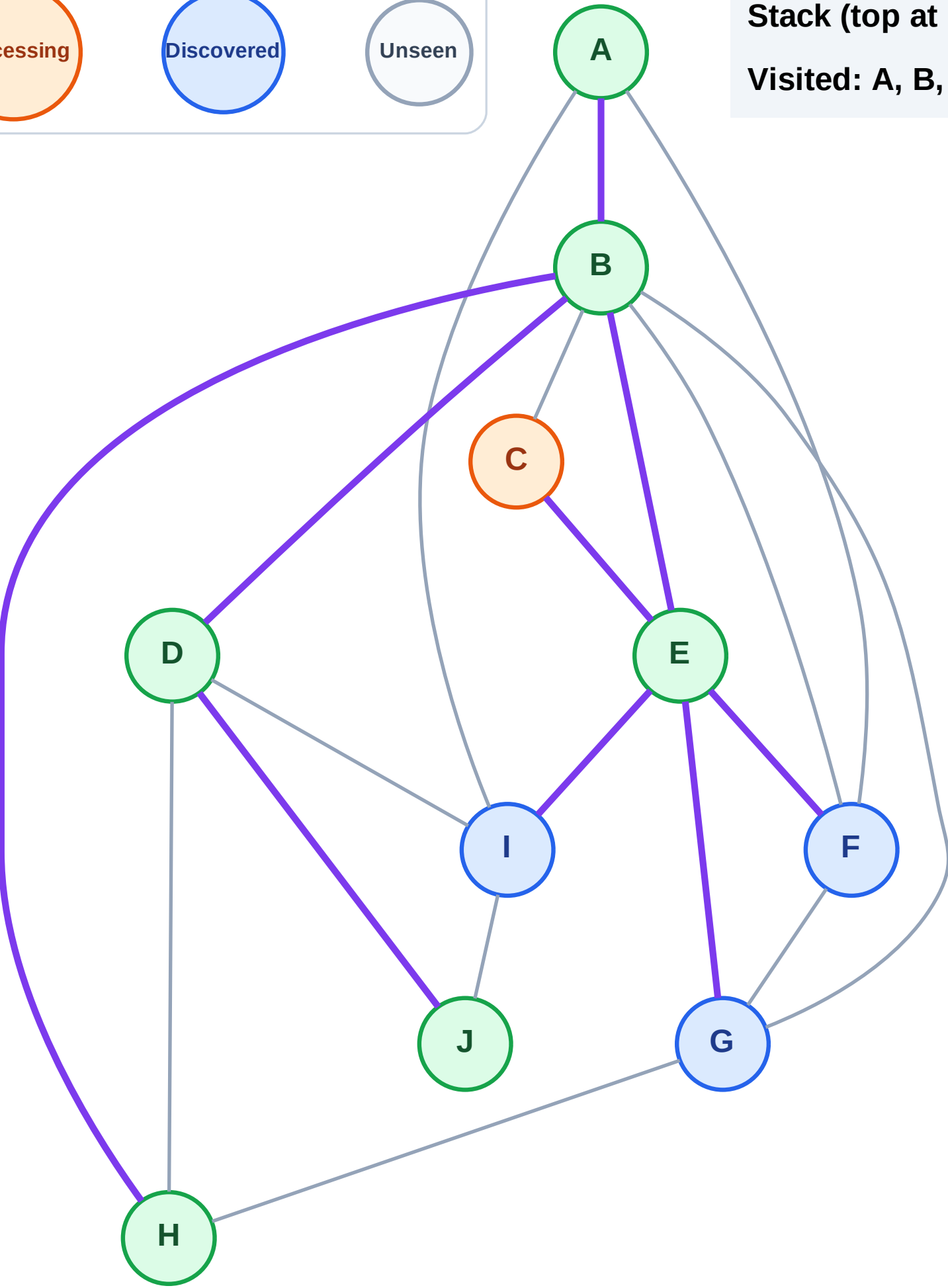
Complete

Processing

Discovered

Unseen

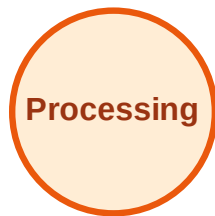
Stack (top at right): I | G | F
Visited: A, B, D, E, H, J



DFS Traversal

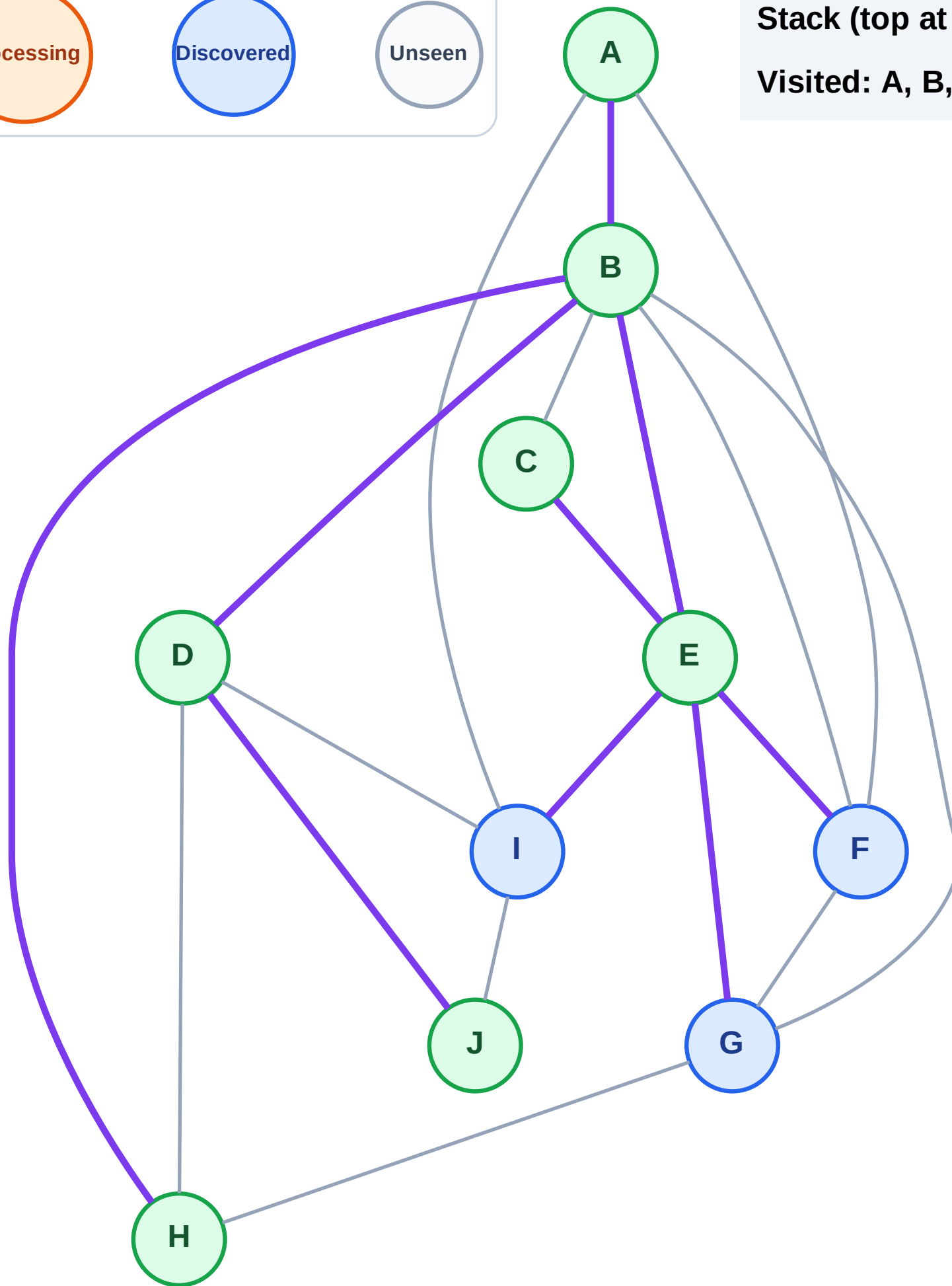
Step 15: No new neighbors from C

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J



DFS Traversal

Step 16: Process node F

Legend

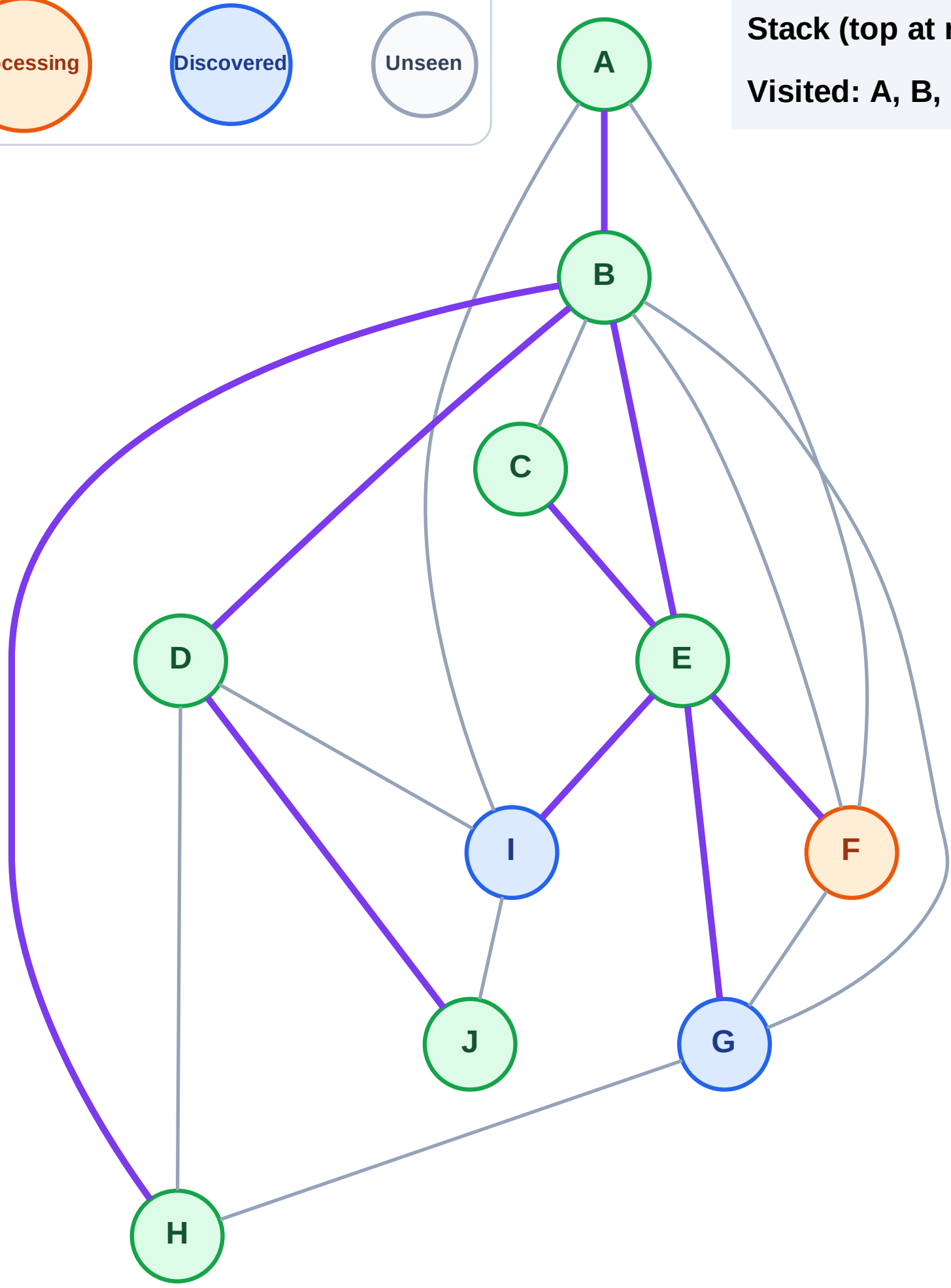
Complete

Processing

Discovered

Unseen

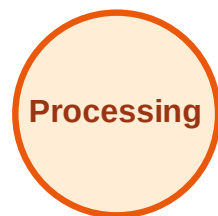
Stack (top at right): I | G
Visited: A, B, C, D, E, H, J



DFS Traversal

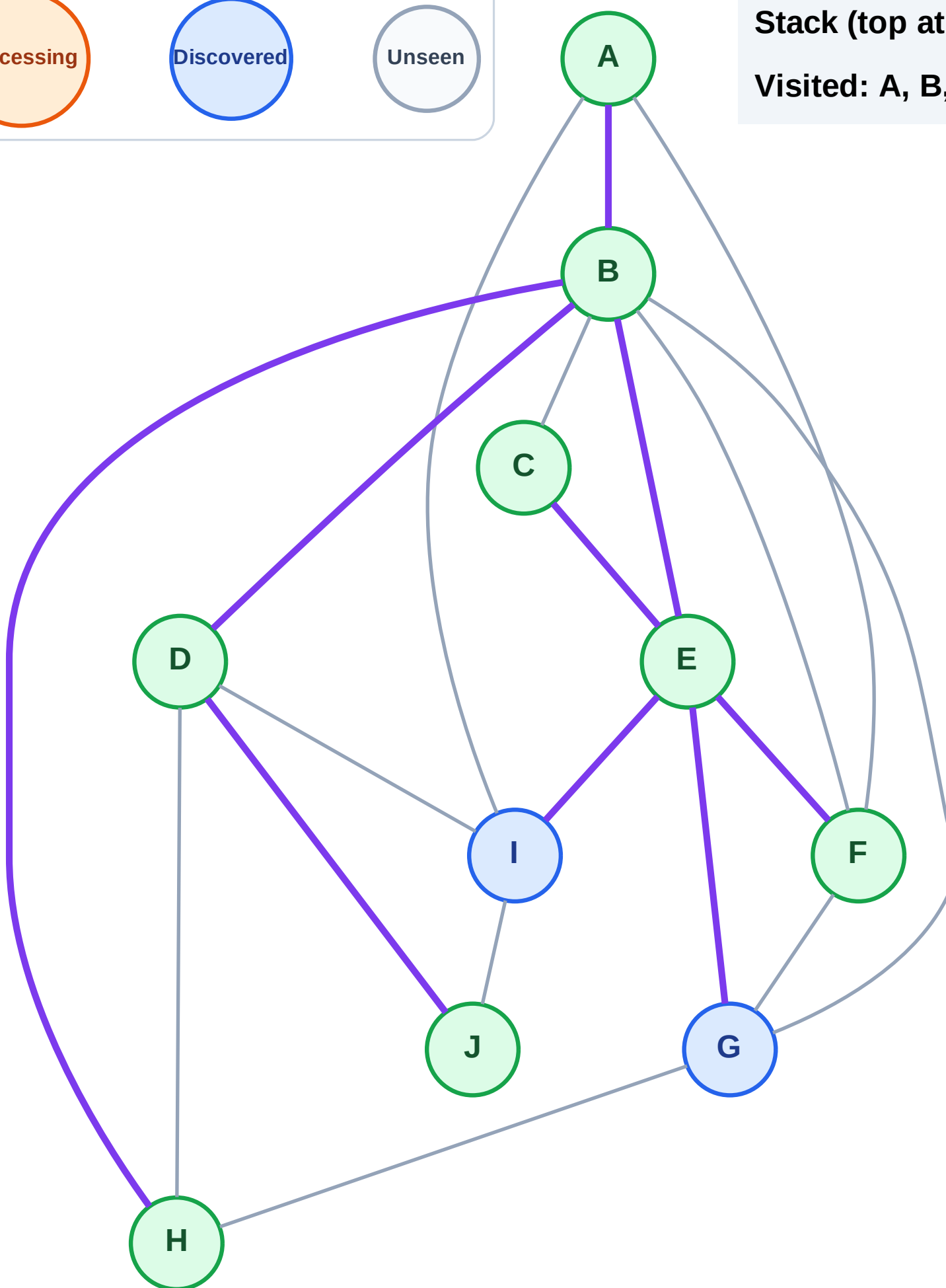
Step 17: No new neighbors from F

Legend



Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 18: Process node G

Legend

Complete

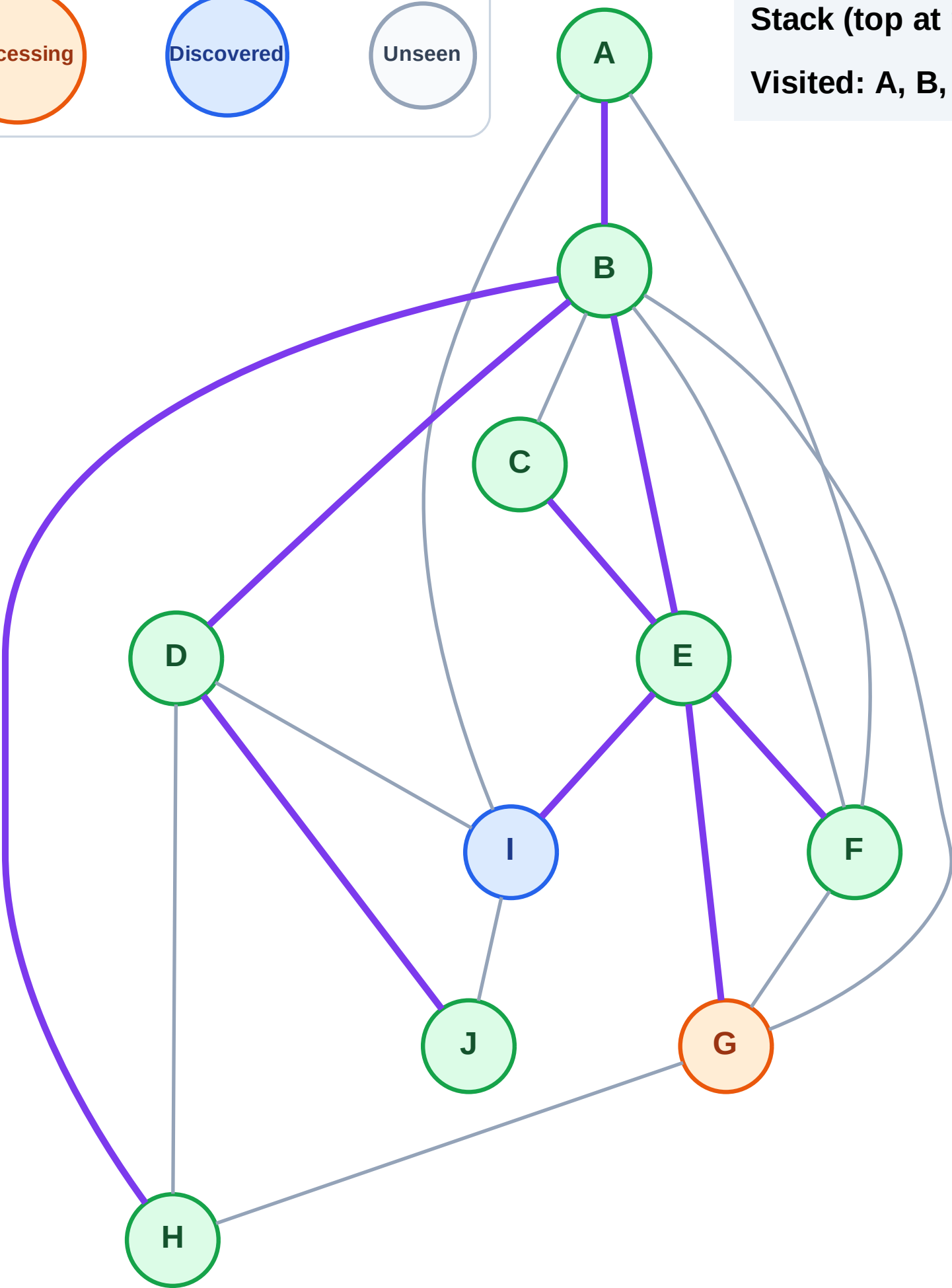
Processing

Discovered

Unseen

Stack (top at right): I

Visited: A, B, C, D, E, F, H, J



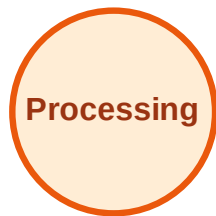
DFS Traversal

Step 19: No new neighbors from G

Legend



Complete



Processing



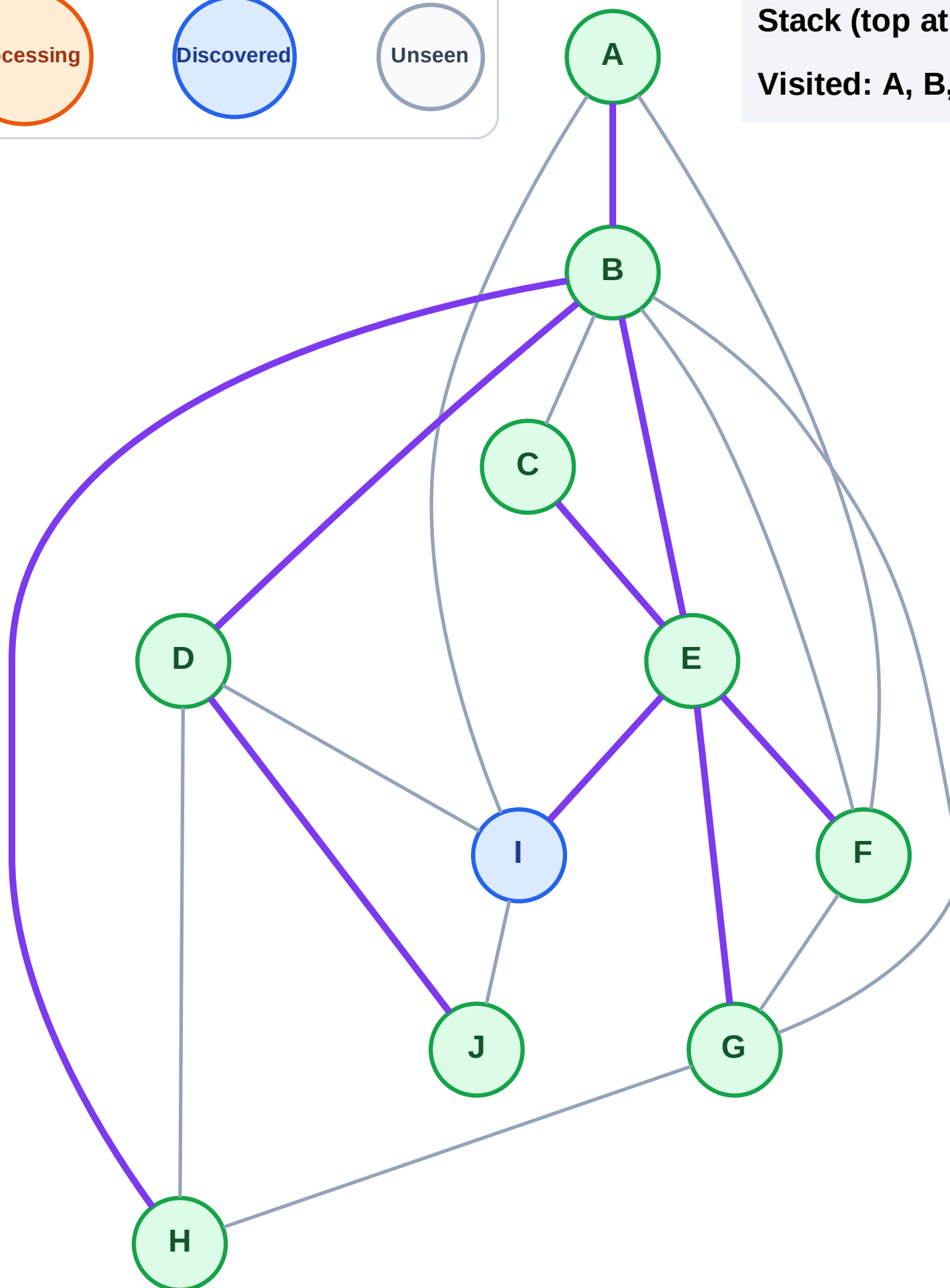
Discovered



Unseen

Stack (top at right): I

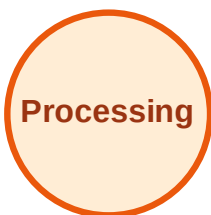
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

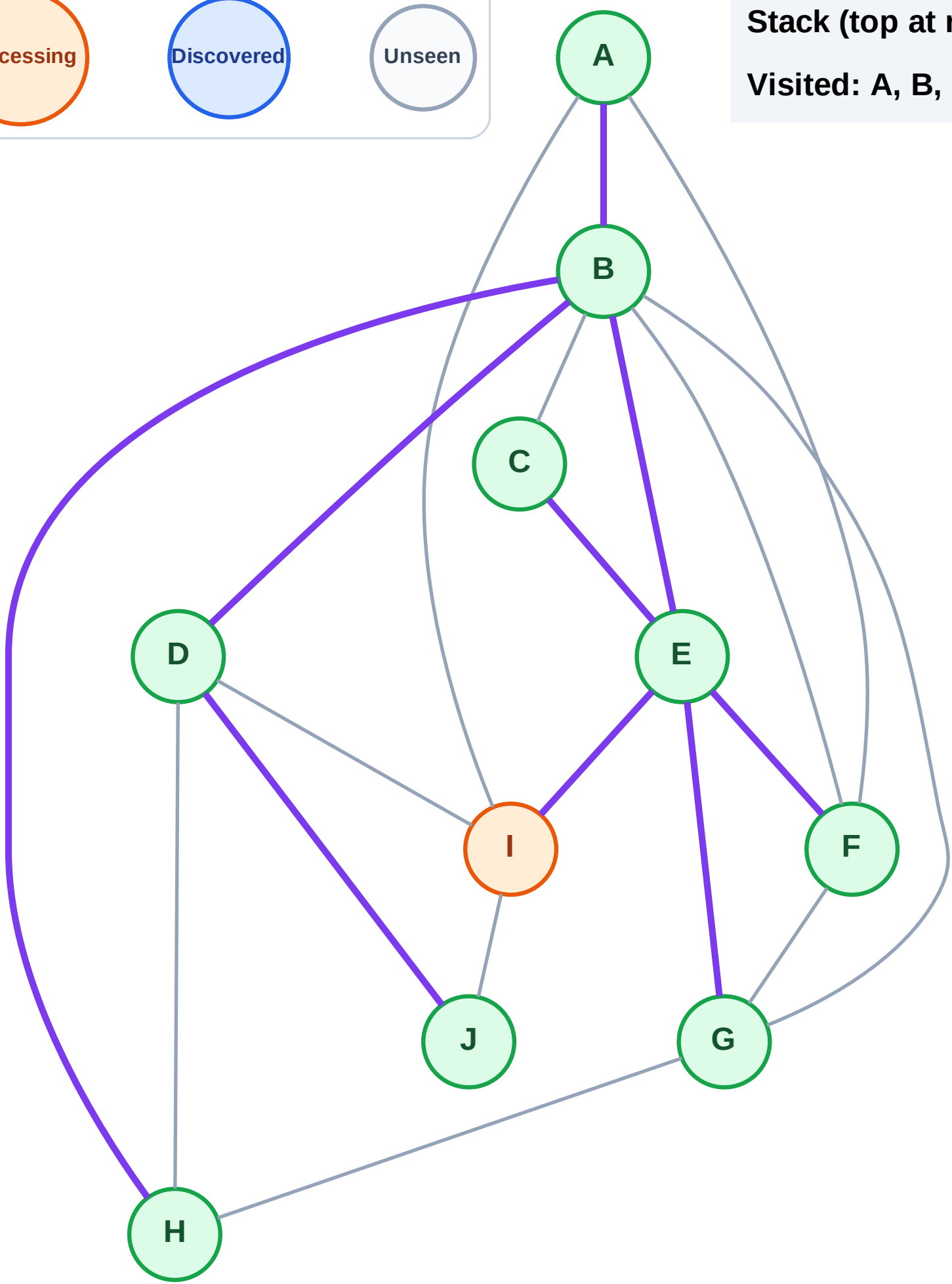
Step 20: Process node I

Legend



Stack (top at right): (empty)

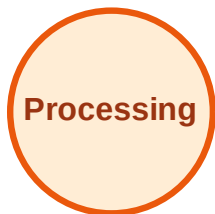
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

